ALTERA Cyclone V SoC Development & Education Board (DE10-Standard)

Title: DE10-Standard Board

Size: B | Document Number: Cover Page | Rev: D

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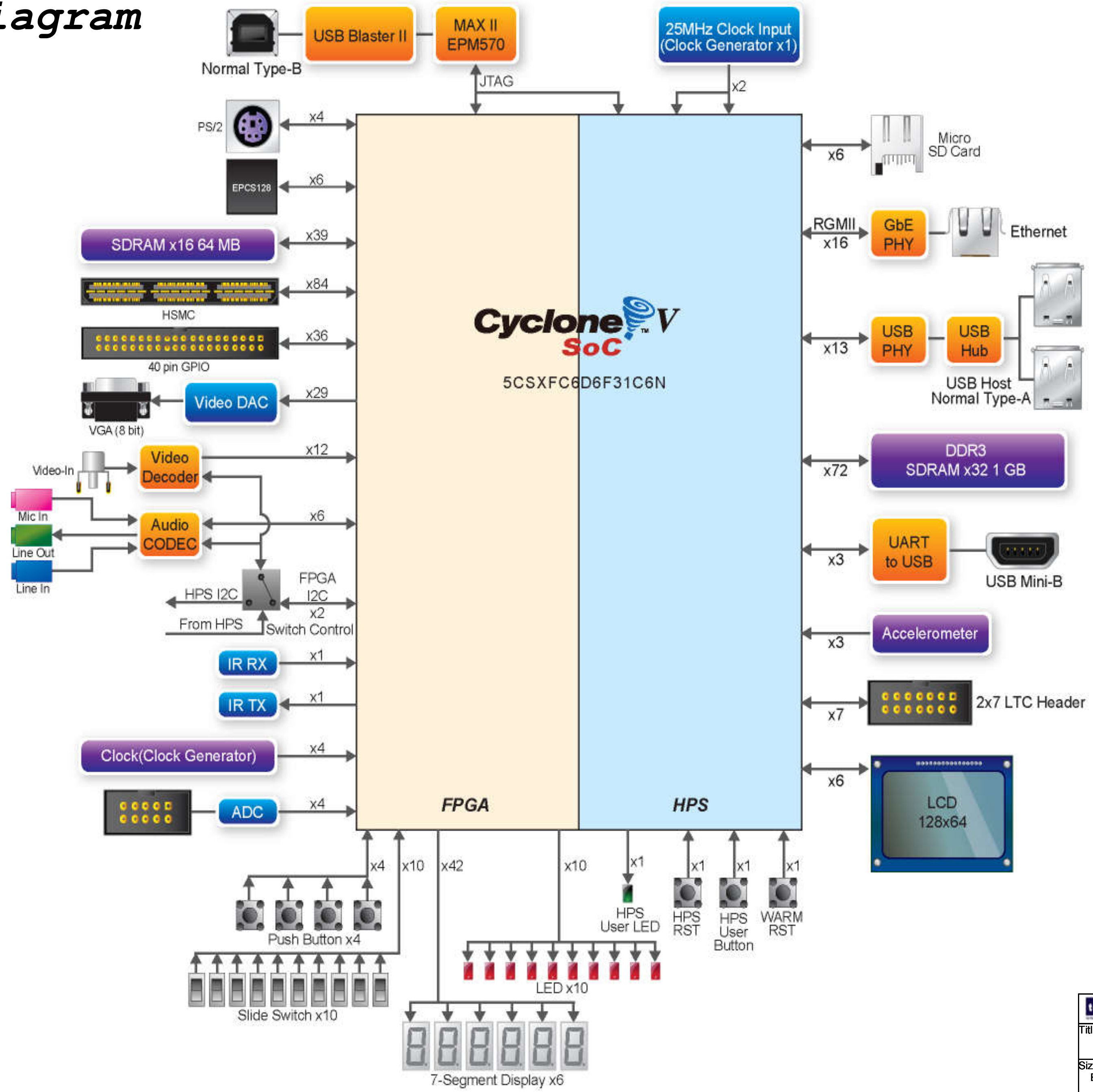
Block Diagram



Title		
DE10-Standard Board		
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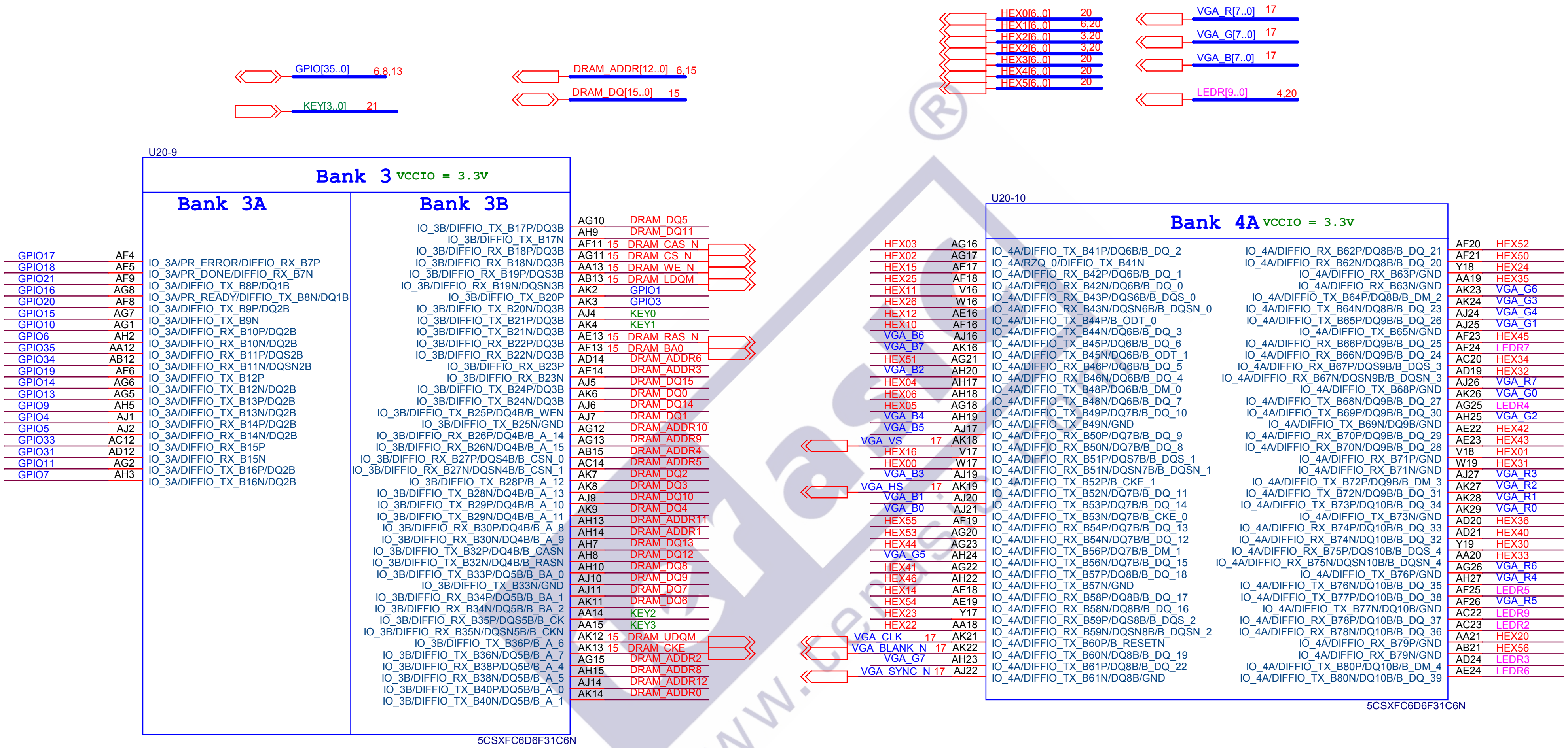
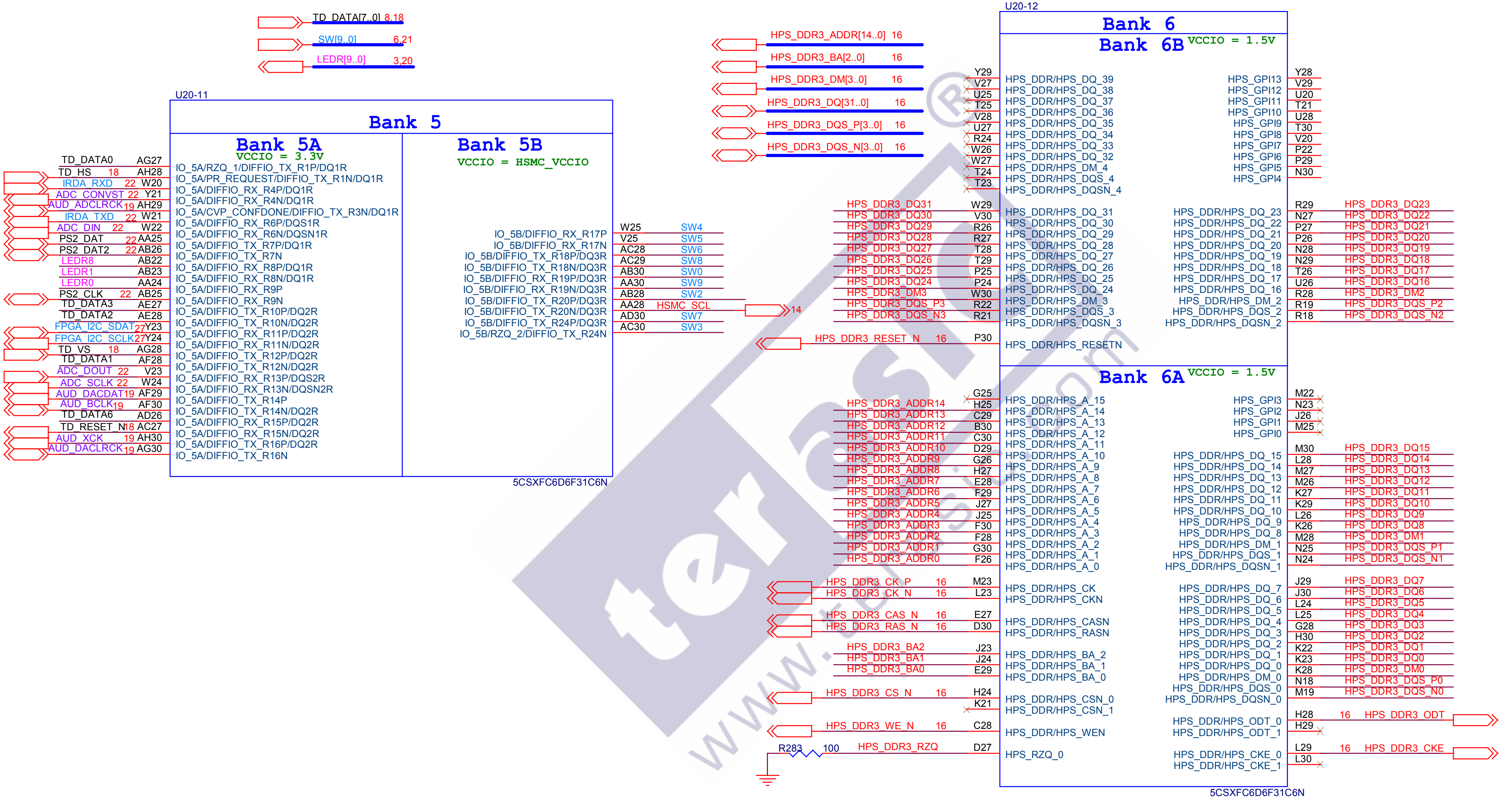


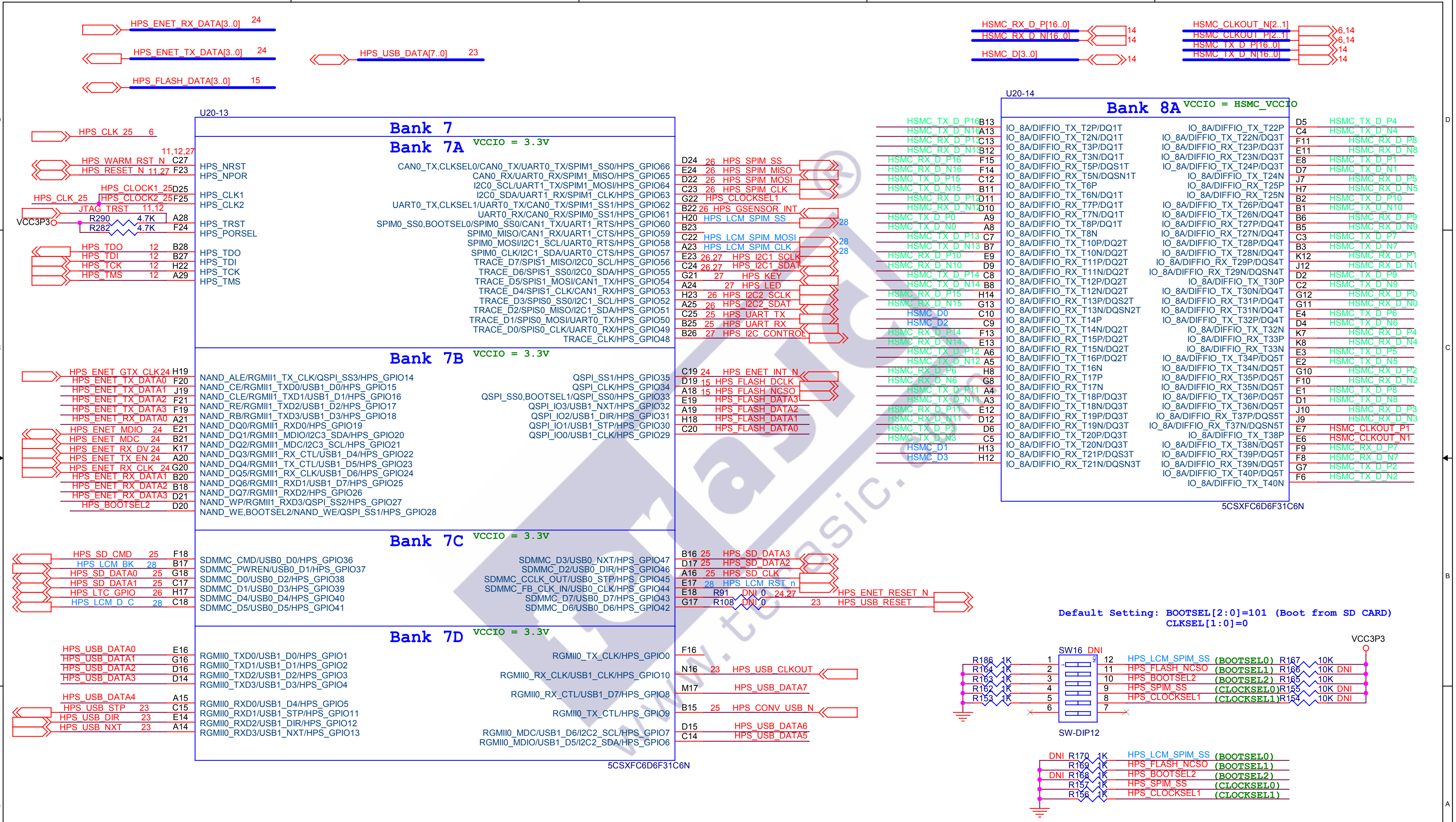
Title		
DE10-Standard Board		
Size: B	Document Number: FPGA BANK 3, BANK 4	Rev: D
Date: Thursday, November 03, 2022	Sheet 3 of 33	



Title: **DE10-Standard Board**

Size	Document Number	Rev
B	FPGA BANK 5, BANK 6	D

Date: Thursday, November 03, 2022 Sheet 4 of 33



Title	DE10-Standard Board	
Size: B	Document Number: FPGA BANK 7, BANK 8	Rev: D
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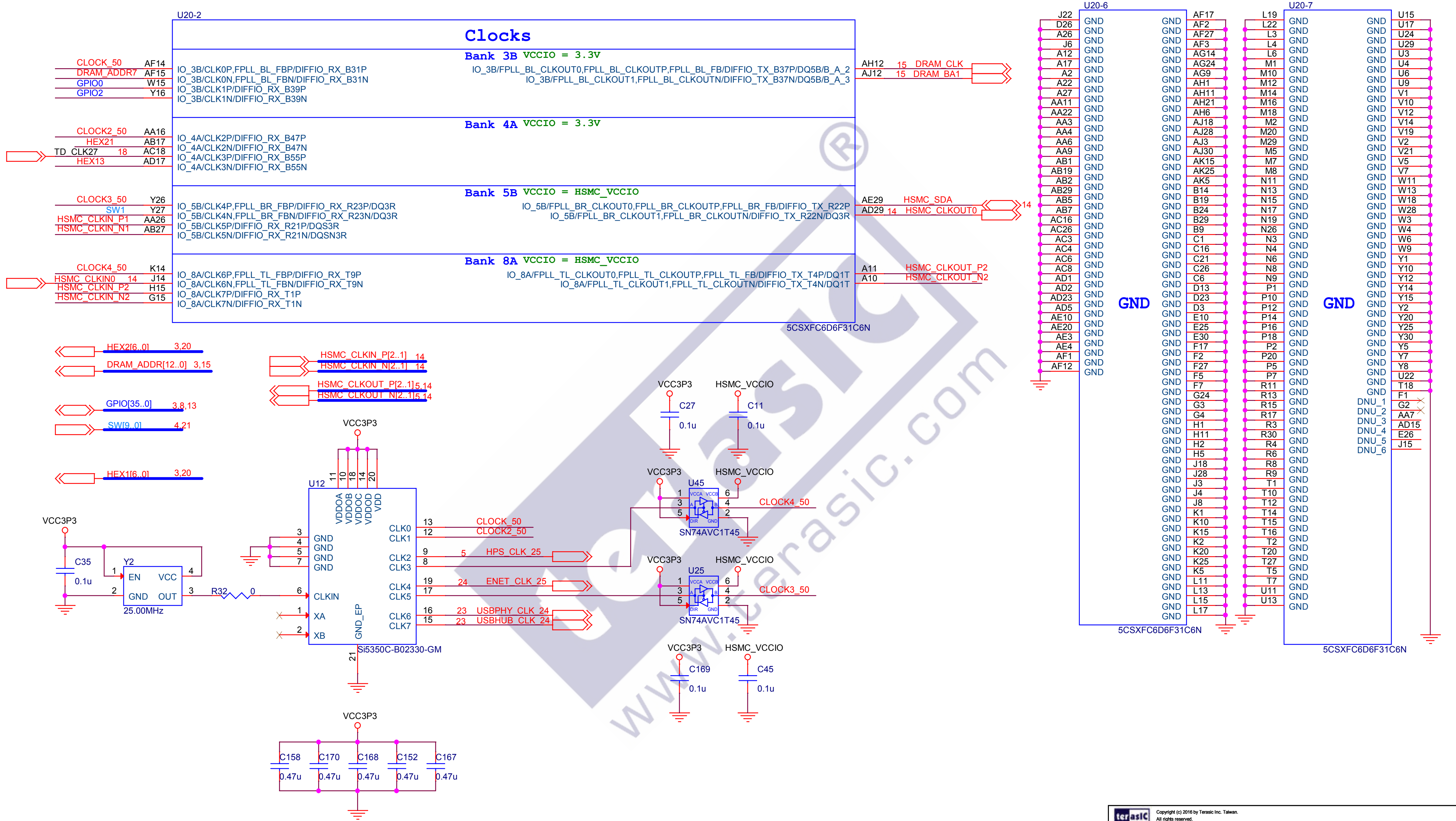
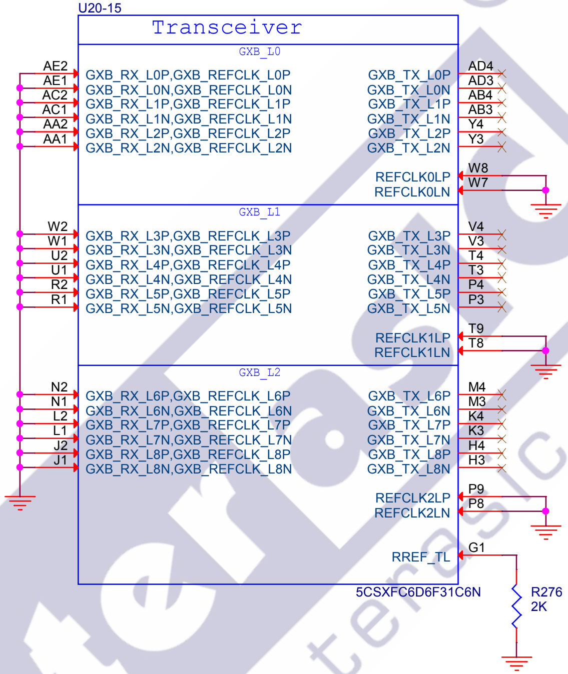


Title

DE10-Standard Board

Size	Document Number	Rev
B	FPGA Clocks & GND	D

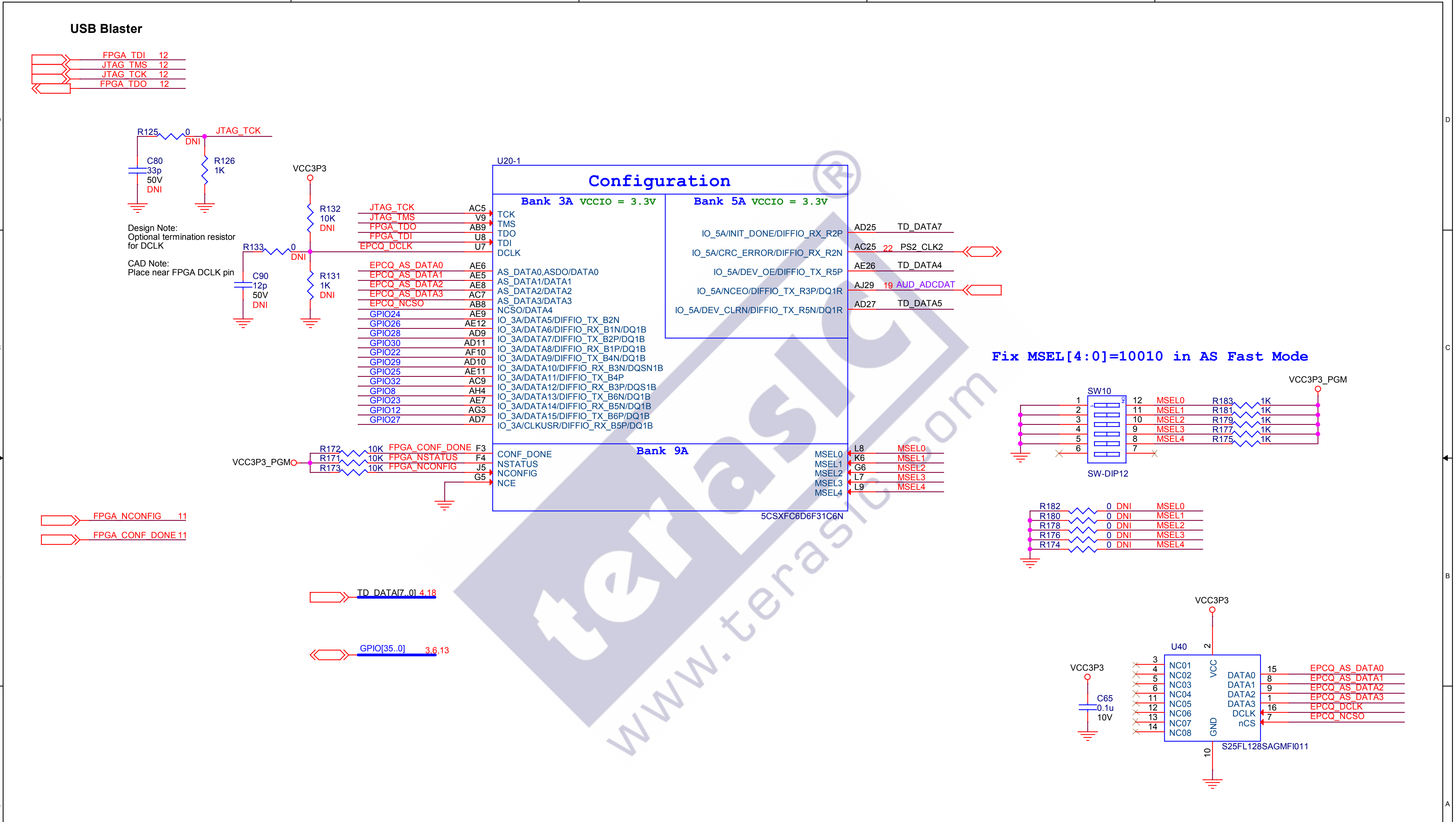
Date: Thursday, November 03, 2022 Sheet 6 of 33



Title
DE10-Standard Board

Size	Document Number	Rev
B	Transceiver	D

Date: Thursday, November 03, 2022 Sheet 7 of 33



Title
DE10-Standard Board

Size	Document Number	Rev
B	FPGA Configuration	D

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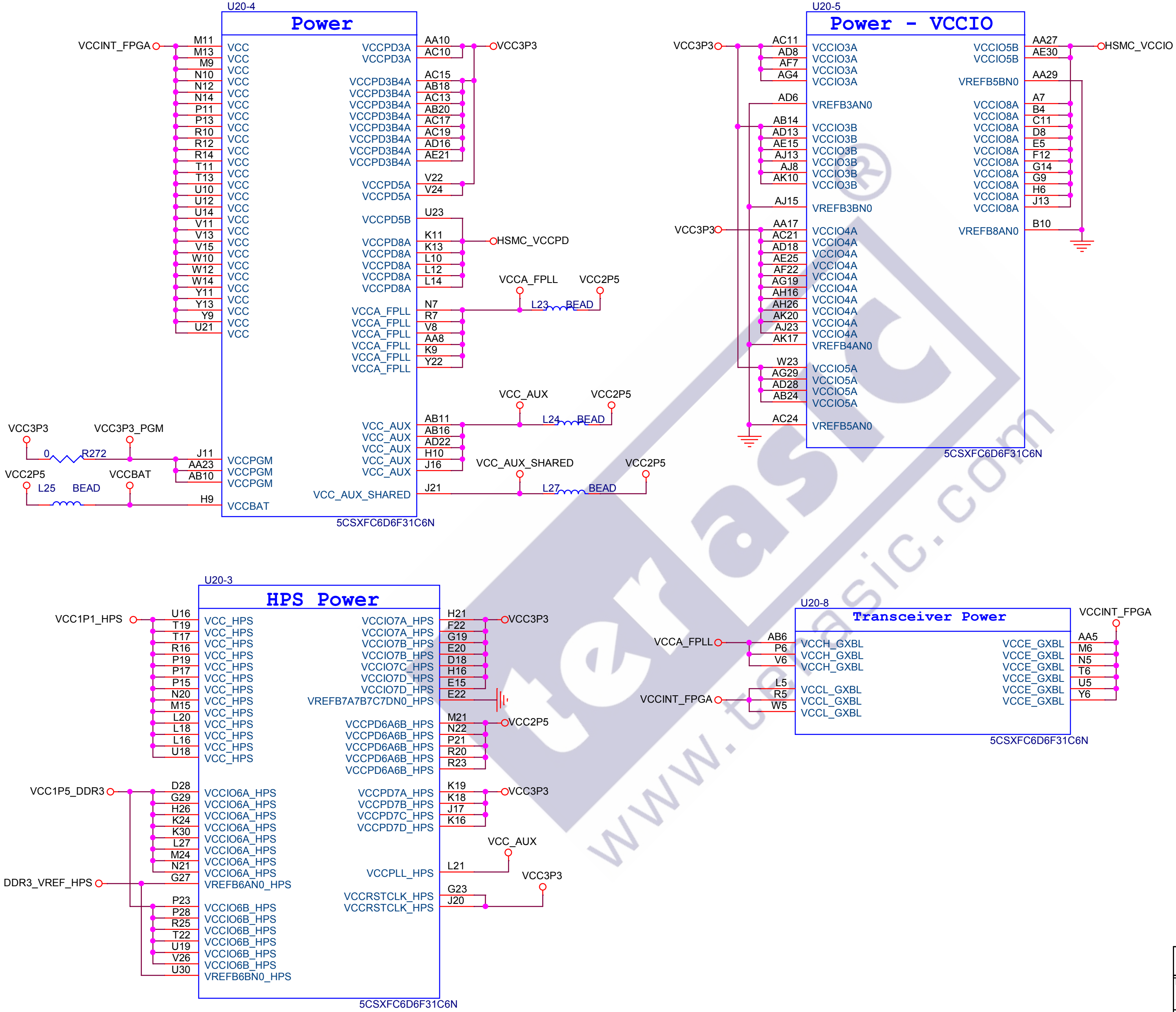
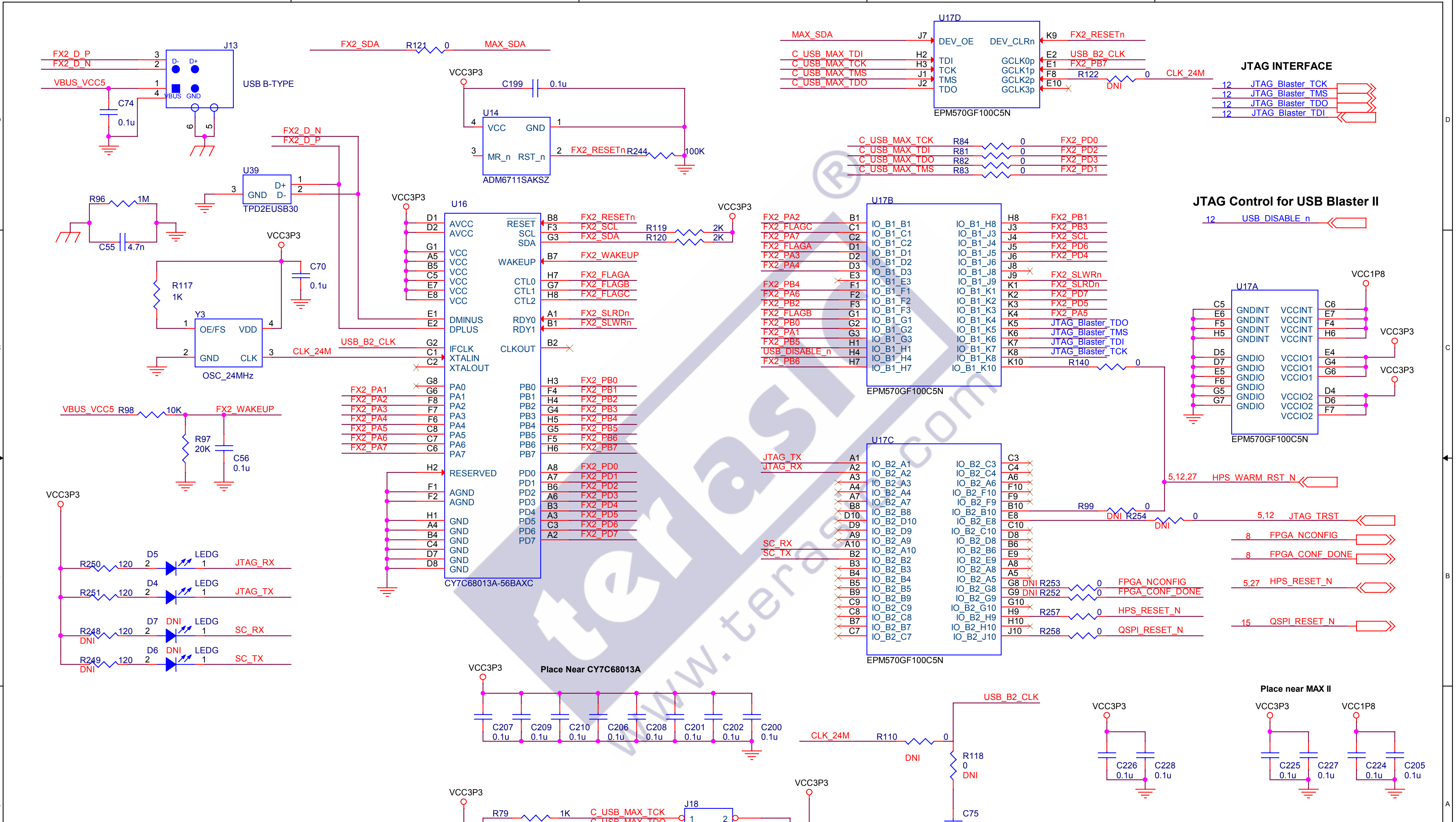
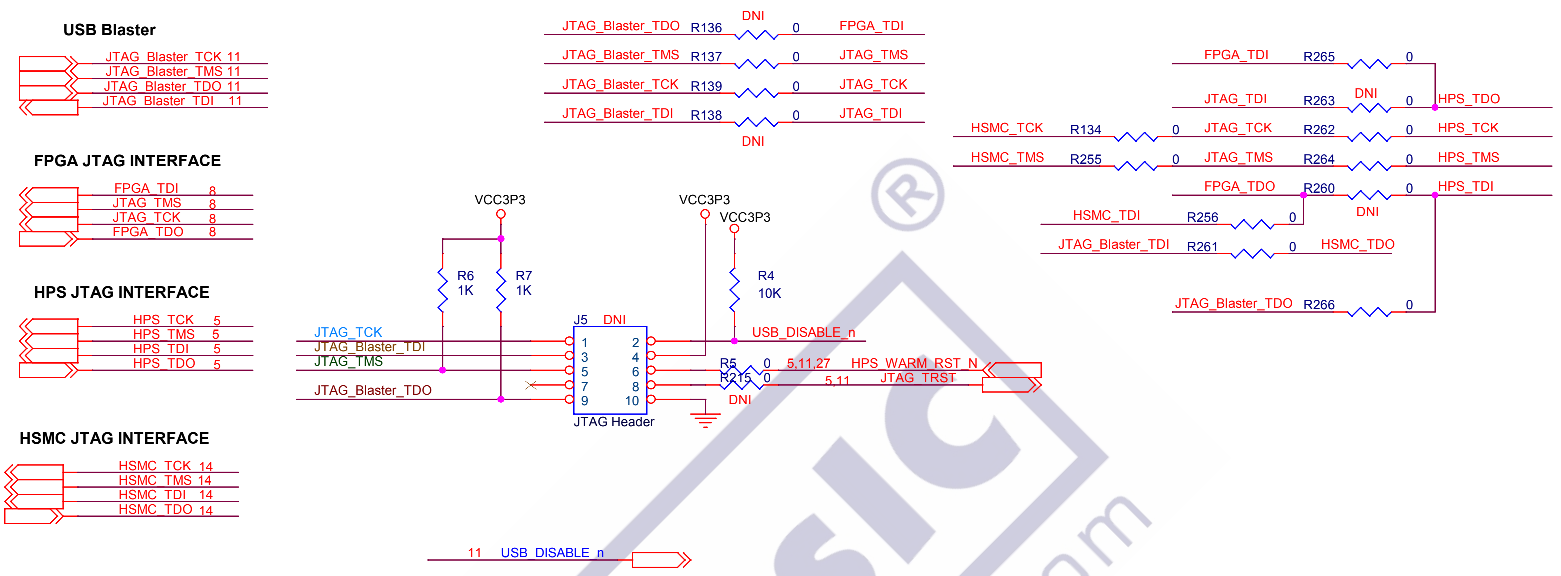
Title		
DE10-Standard Board		
Size: B	Document Number: FPGA Decoupling	Rev: D
Date: Thursday, November 03, 2022	Sheet 9 of 33	



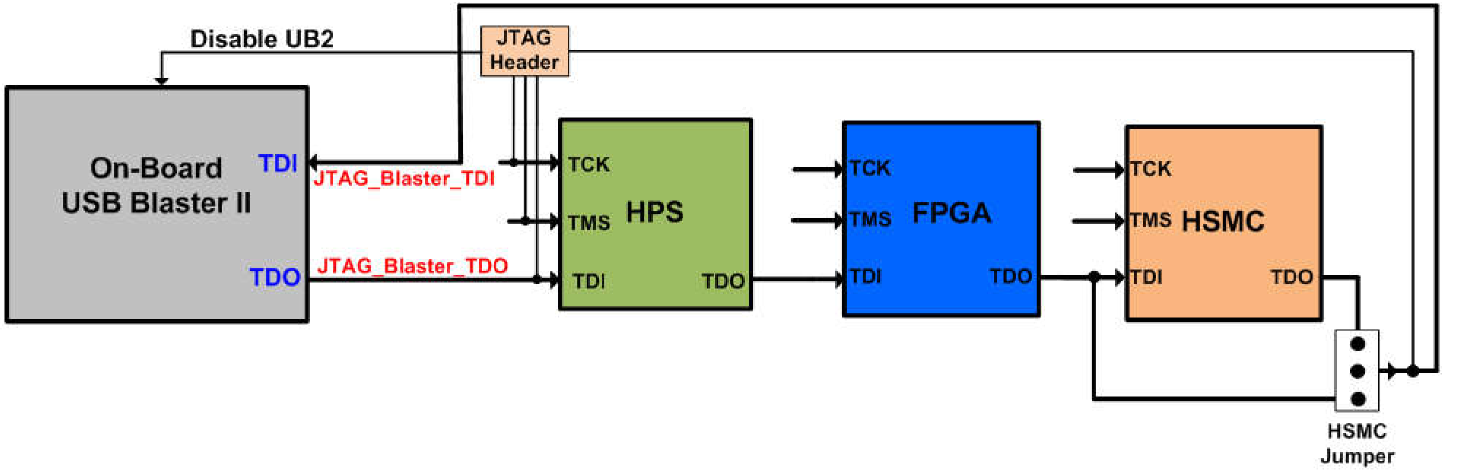
Title	DE10-Standard Board	
Size B	Document Number FPGA Power	Rev D
Date: Thursday, November 03, 2022	Sheet 10 of 33	



Title	DE10-Standard Board	
Size B	Document Number USB Blaster II	Rev D
Date: Thursday, November 03, 2022	Sheet 11 of 33	



JTAG Chain



Title: DE10-Standard Board		
Size: B	Document Number: JTAG Chain	Rev: D
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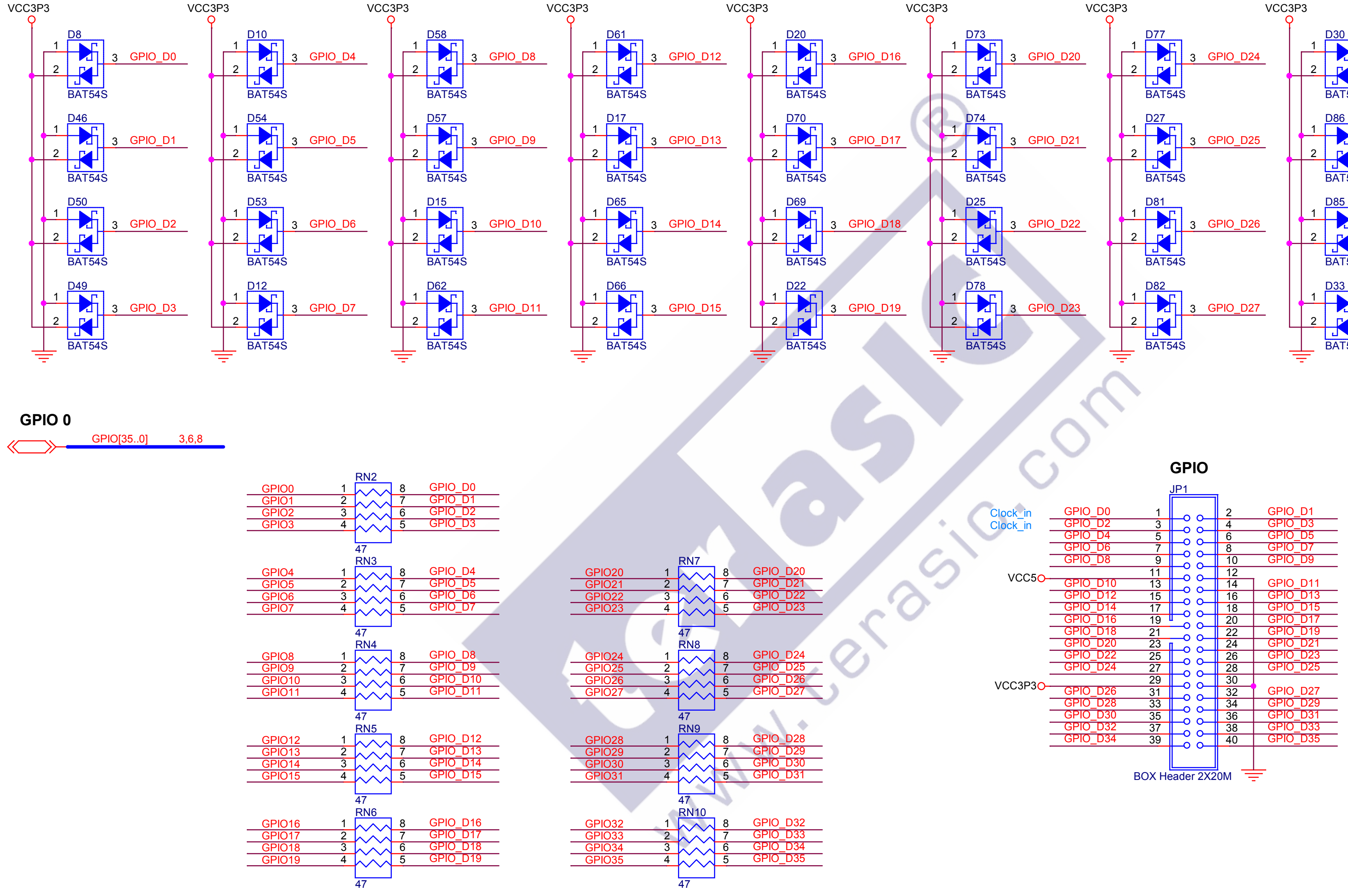
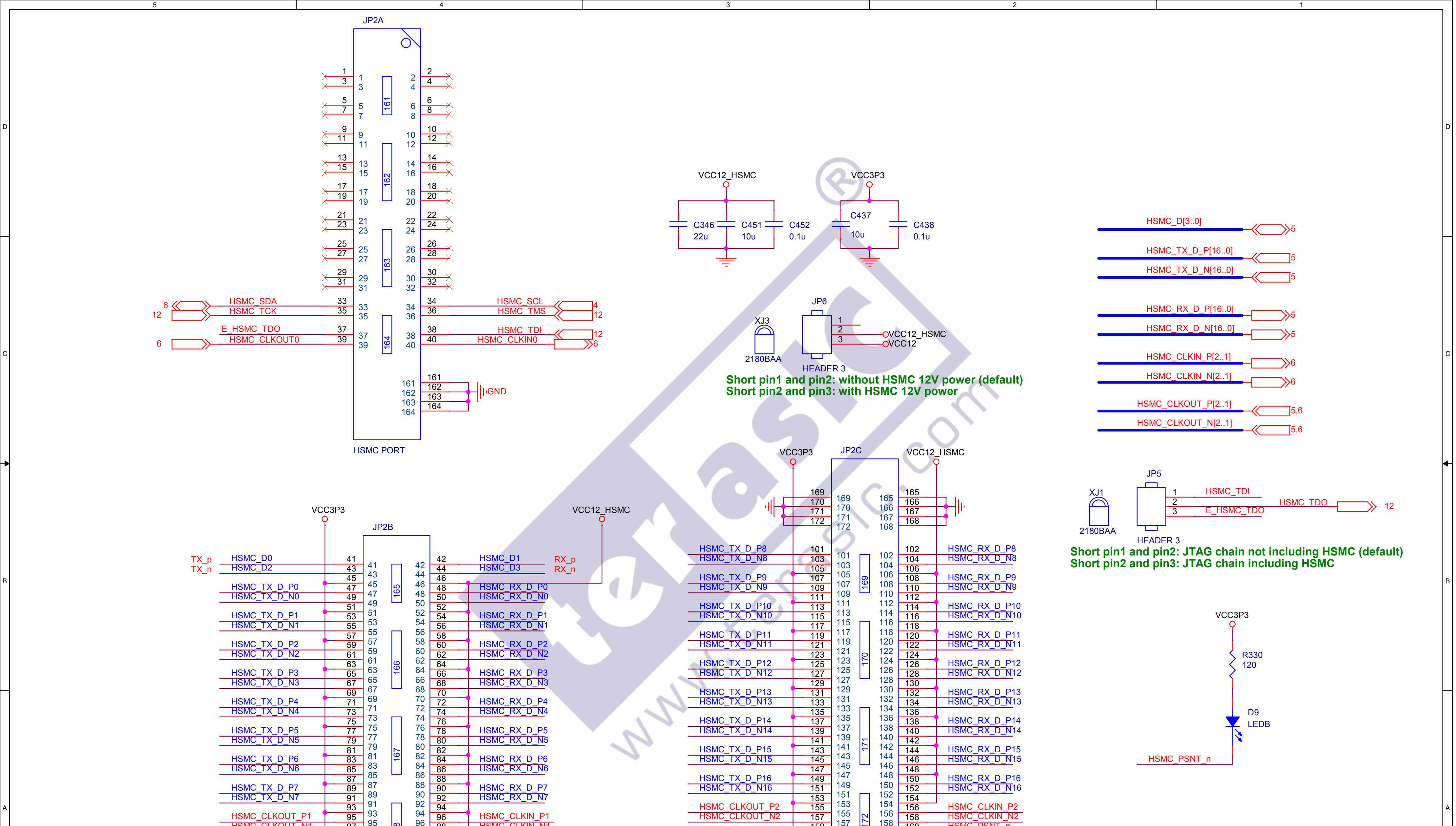


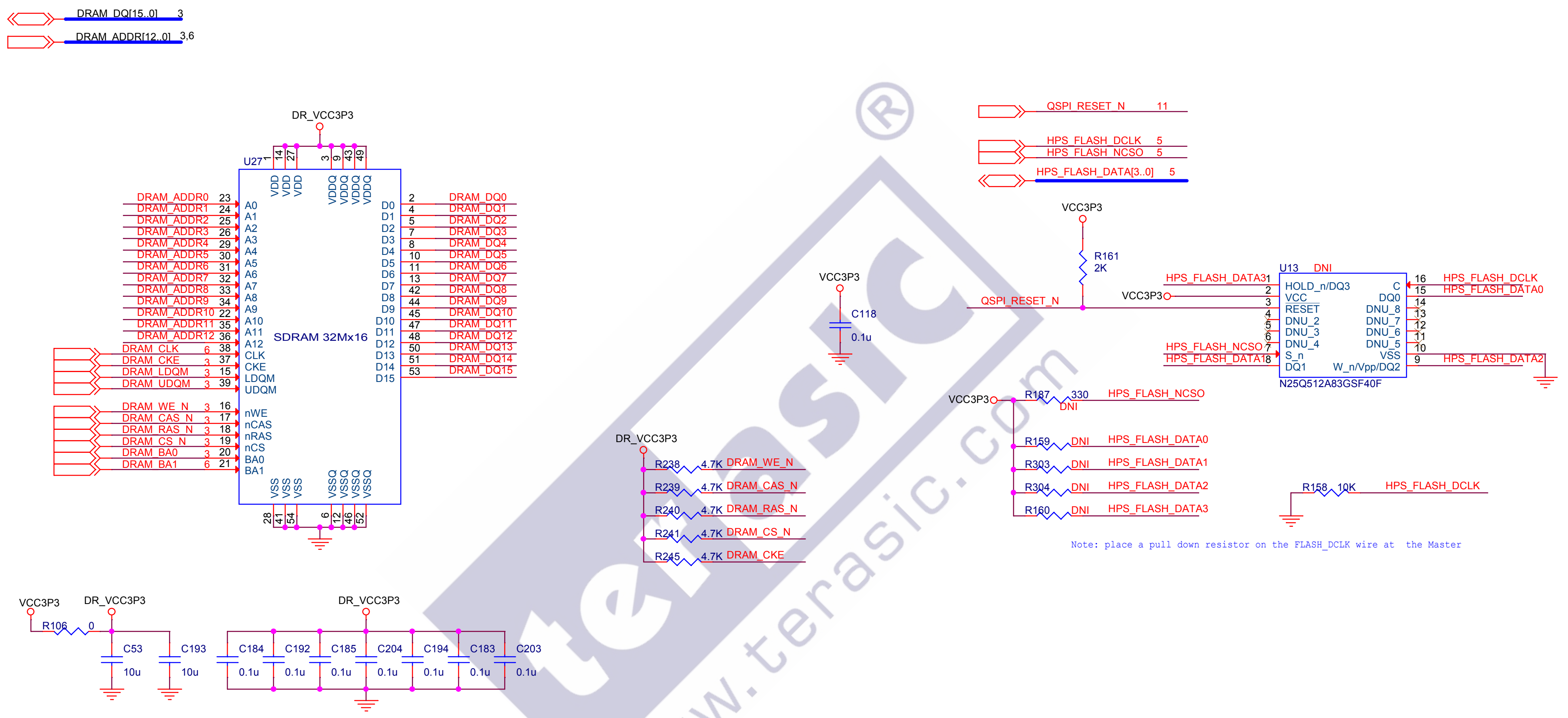
Title
DE10-Standard Board

Size	Document Number	Rev
B	GPIO	D

Date: Thursday, November 03, 2022 Sheet 13 of 33



Title: DE10-Standard Board		
Size: B	Document Number: HSMC	Rev: D
Date: Thursday, November 03, 2022	Sheet 14 of 33	



Title: **DE10-Standard Board**

Size	Document Number	Rev
B	SDRAM & HPS QSPI Flash	D

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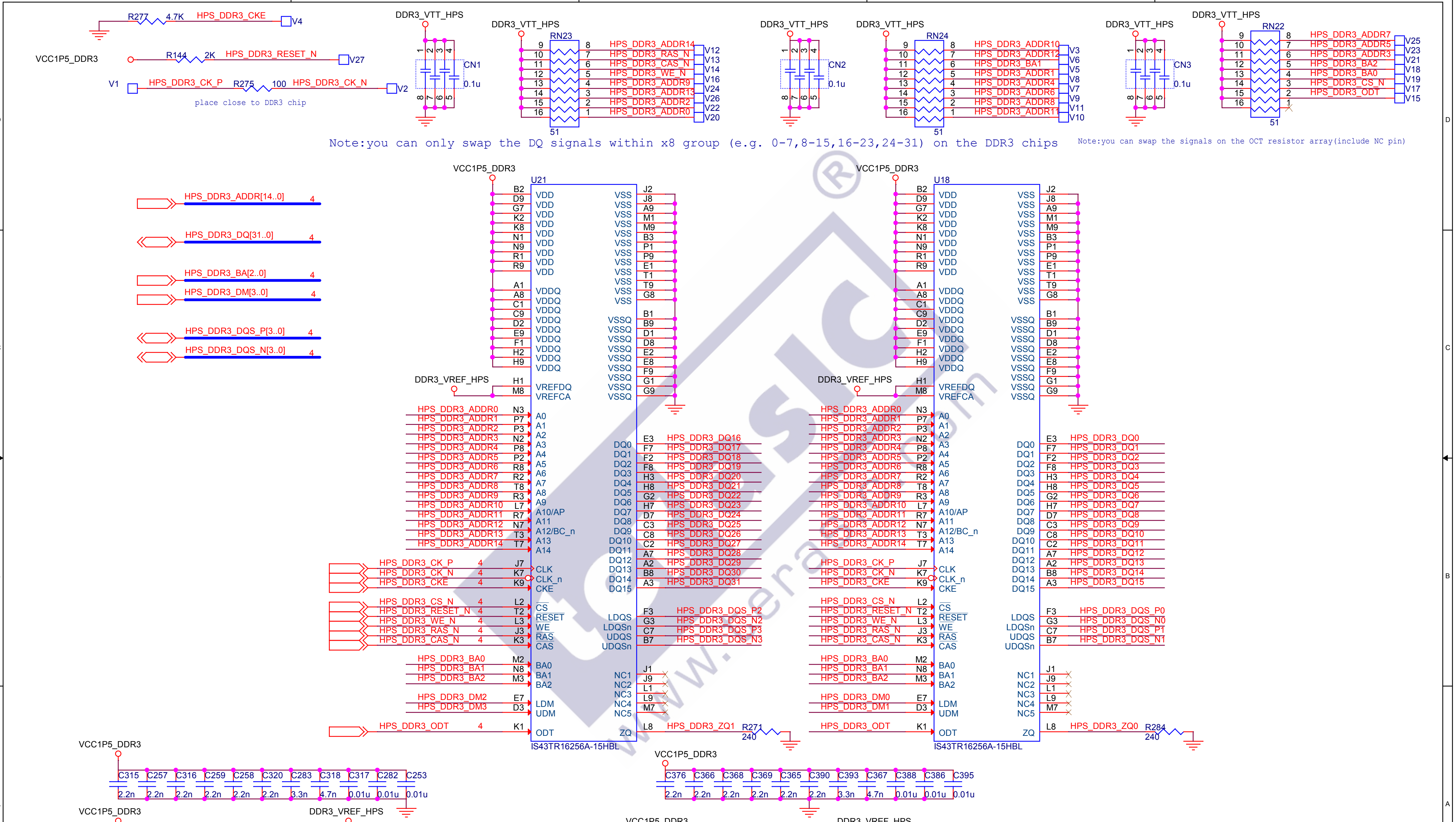

Title: **DE10-Standard Board**

Size: B | Document Number: HPS DDR3 SDRAM | Rev: D

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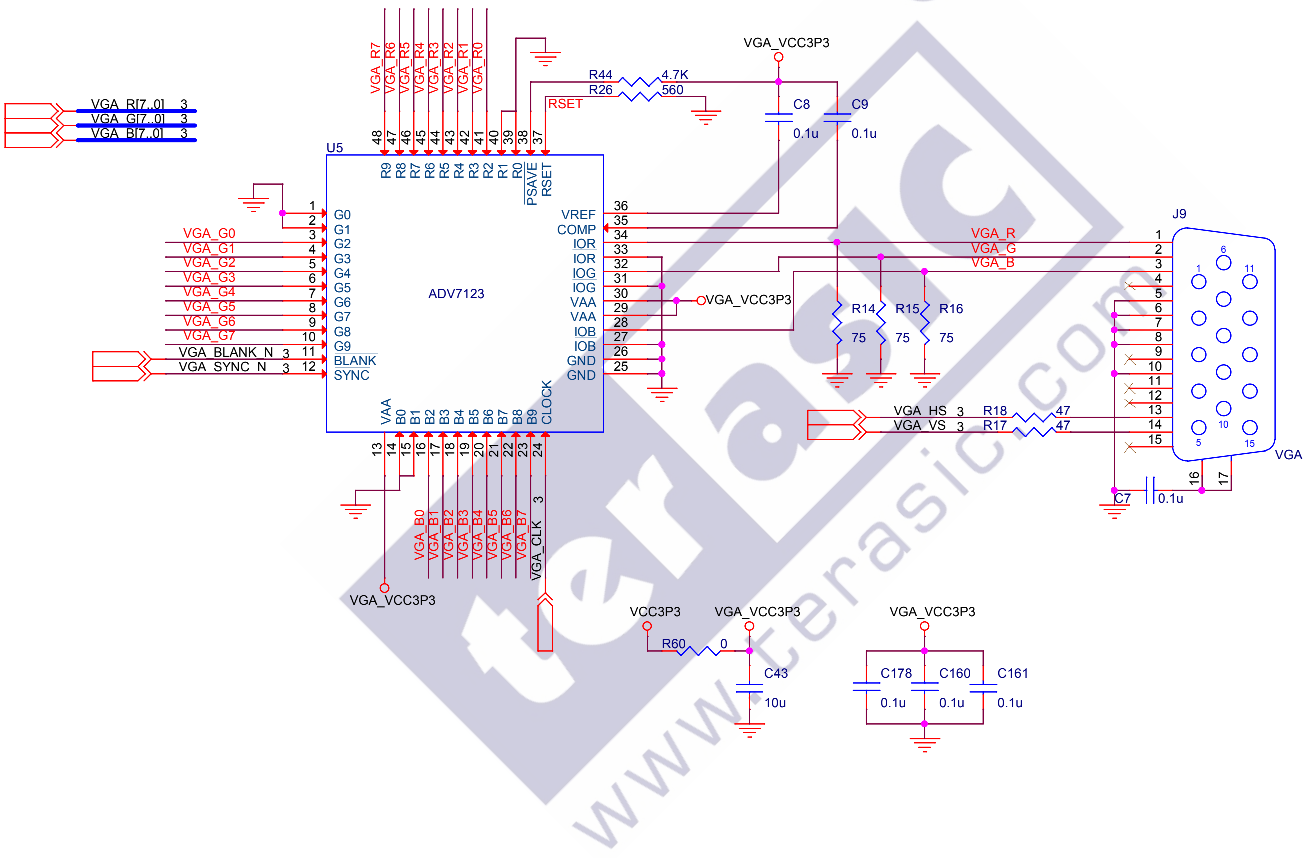
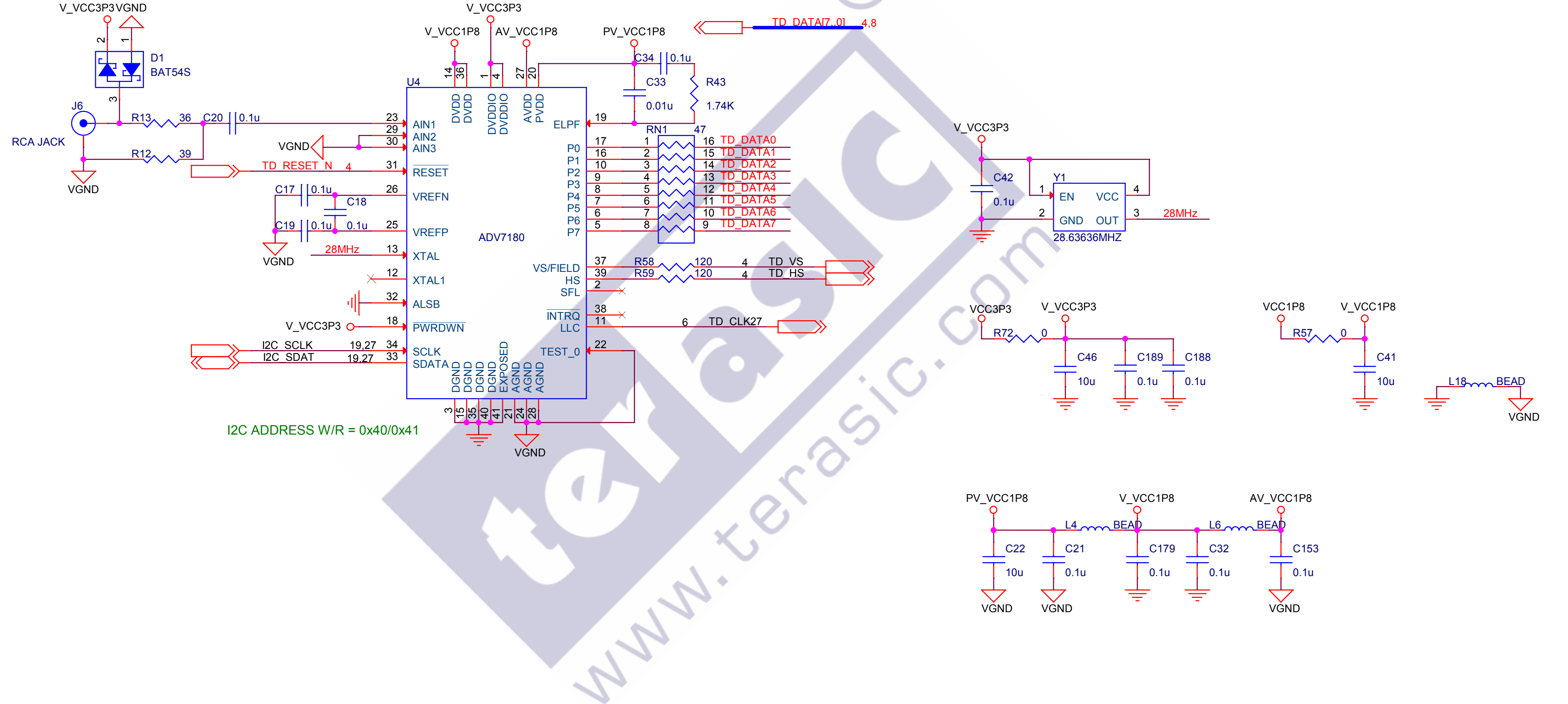


Title
DE10-Standard Board

Size	Document Number	Rev
B	ADV7123 VGA	D

Date: Thursday, November 03, 2022 Sheet 17 of 33



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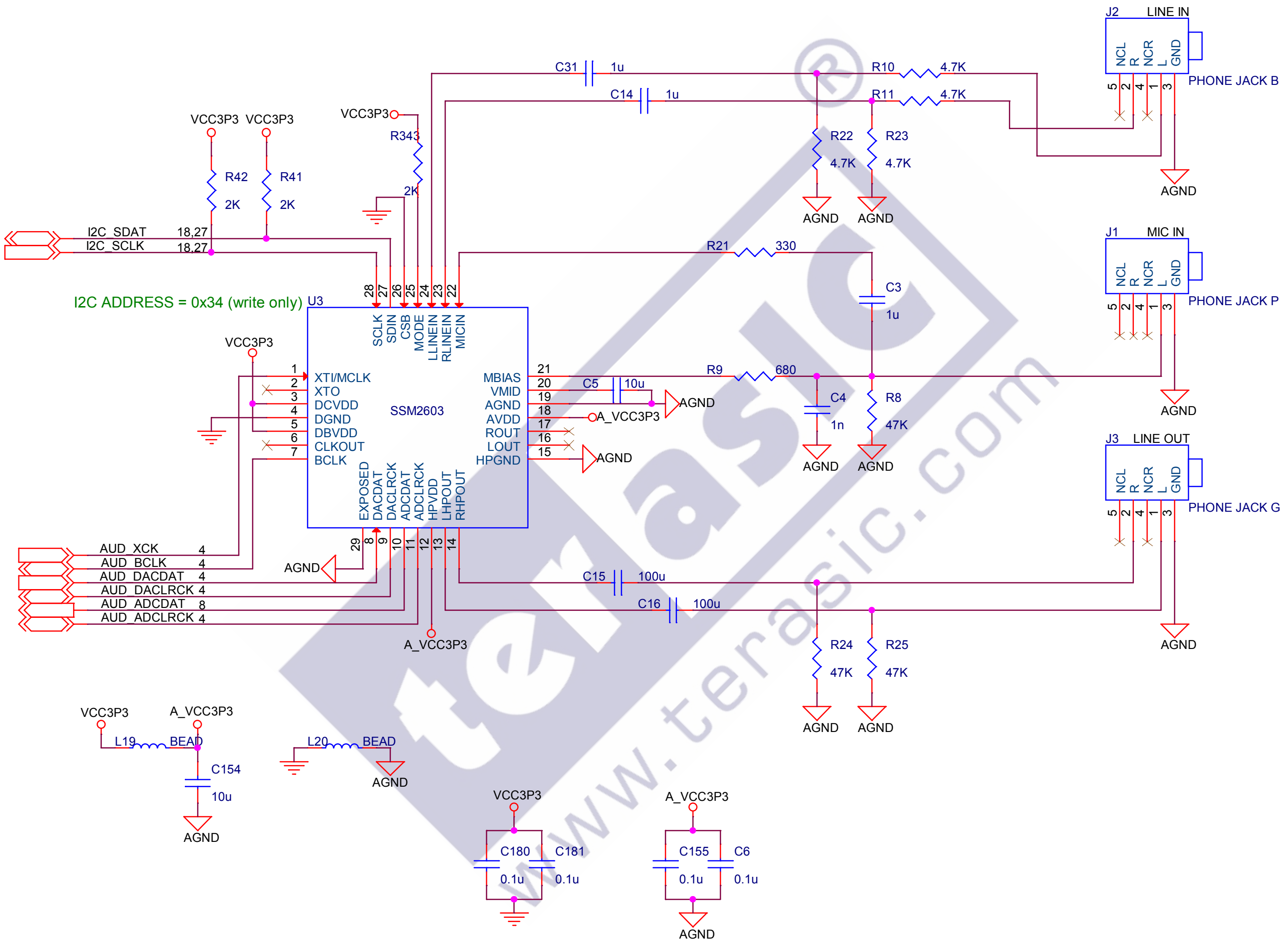
Title
DE10-Standard Board

Size	Document Number	Rev
B	ADV7180 Video Decoder	D

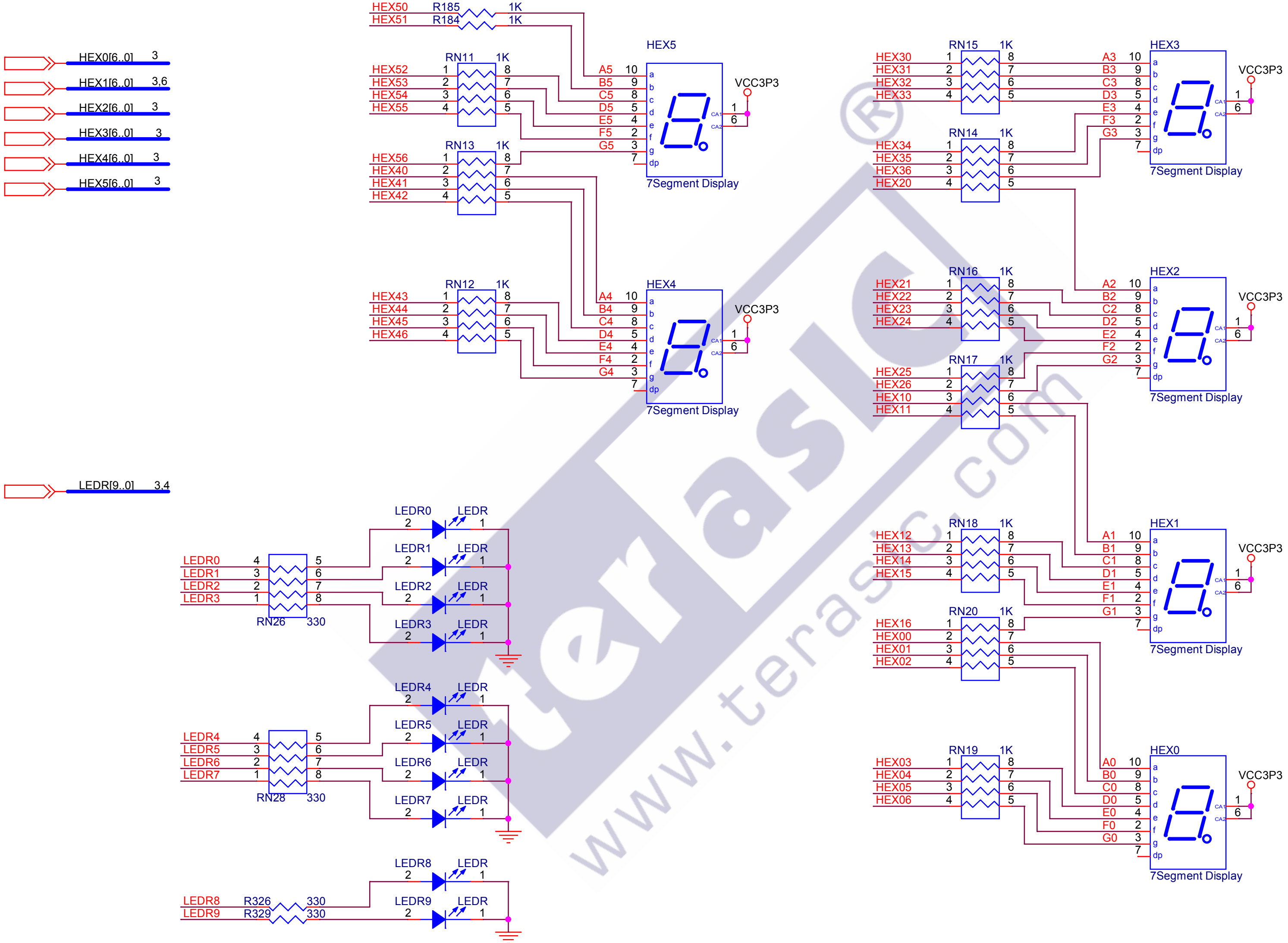
Date: Thursday, November 03, 2022 Sheet 18 of 33



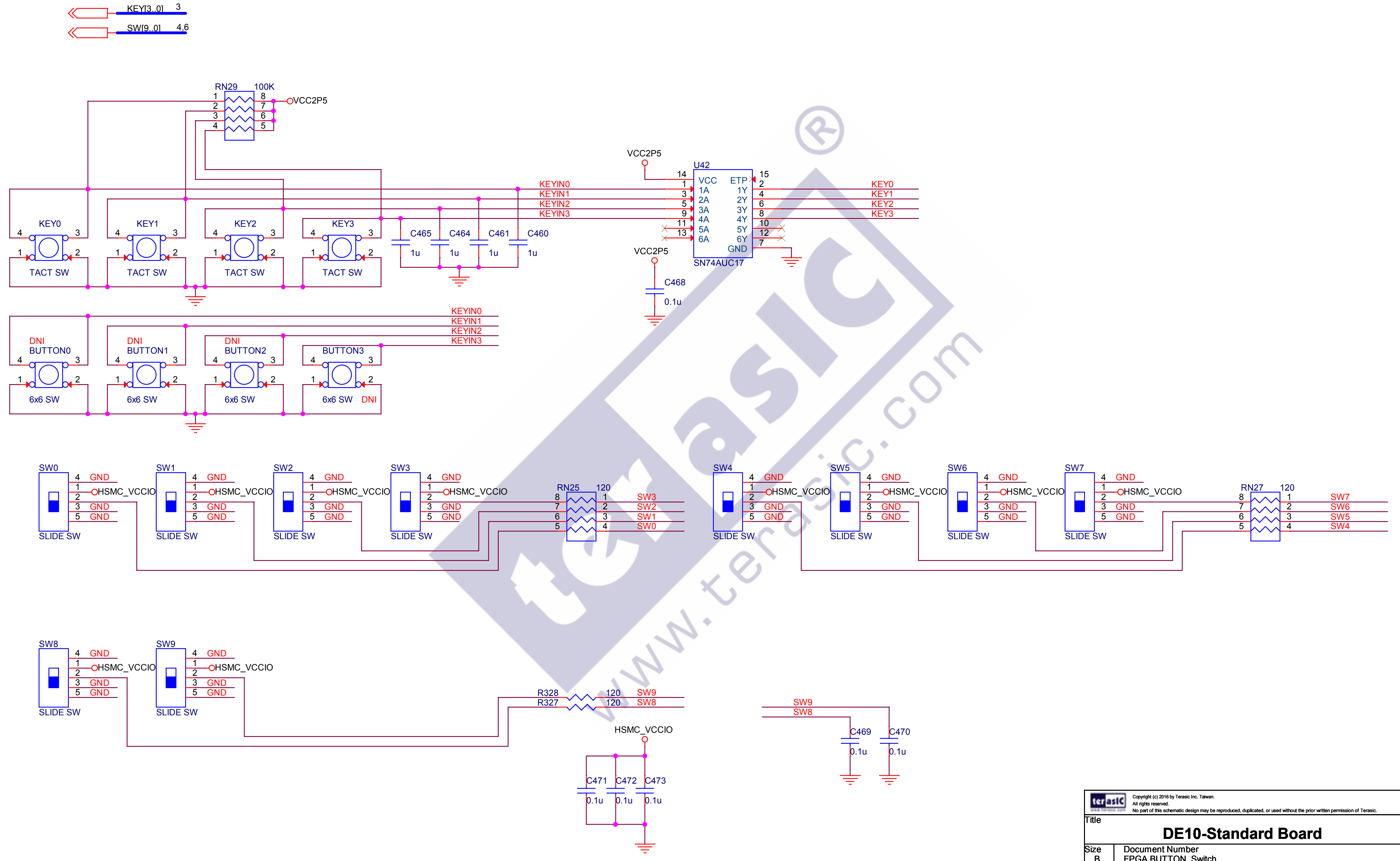
Title
DE10-Standard Board

Size B | Document Number: Audio CODEC | Rev D

Date: Thursday, November 03, 2022 | Sheet 19 of 33



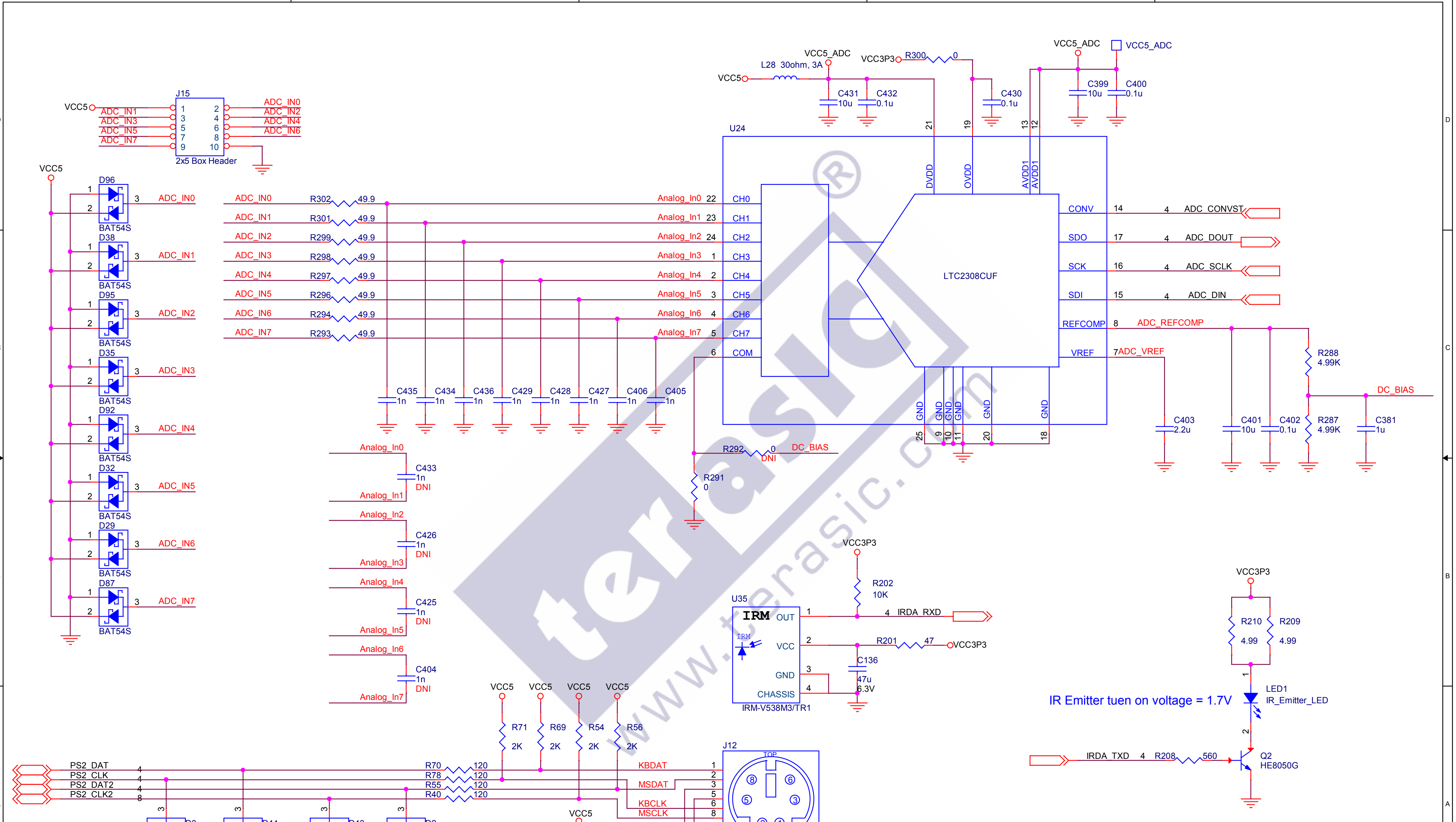
Title		
DE10-Standard Board		
Size: B	Document Number: 7-Segment Display, LED	Rev: D
Date: Thursday, November 03, 2022	Sheet 20 of 33	



Title
DE10-Standard Board

Size	Document Number	Rev
B	FPGA BUTTON, Switch	D

Date: Thursday, November 03, 2022 Sheet 21 of 33



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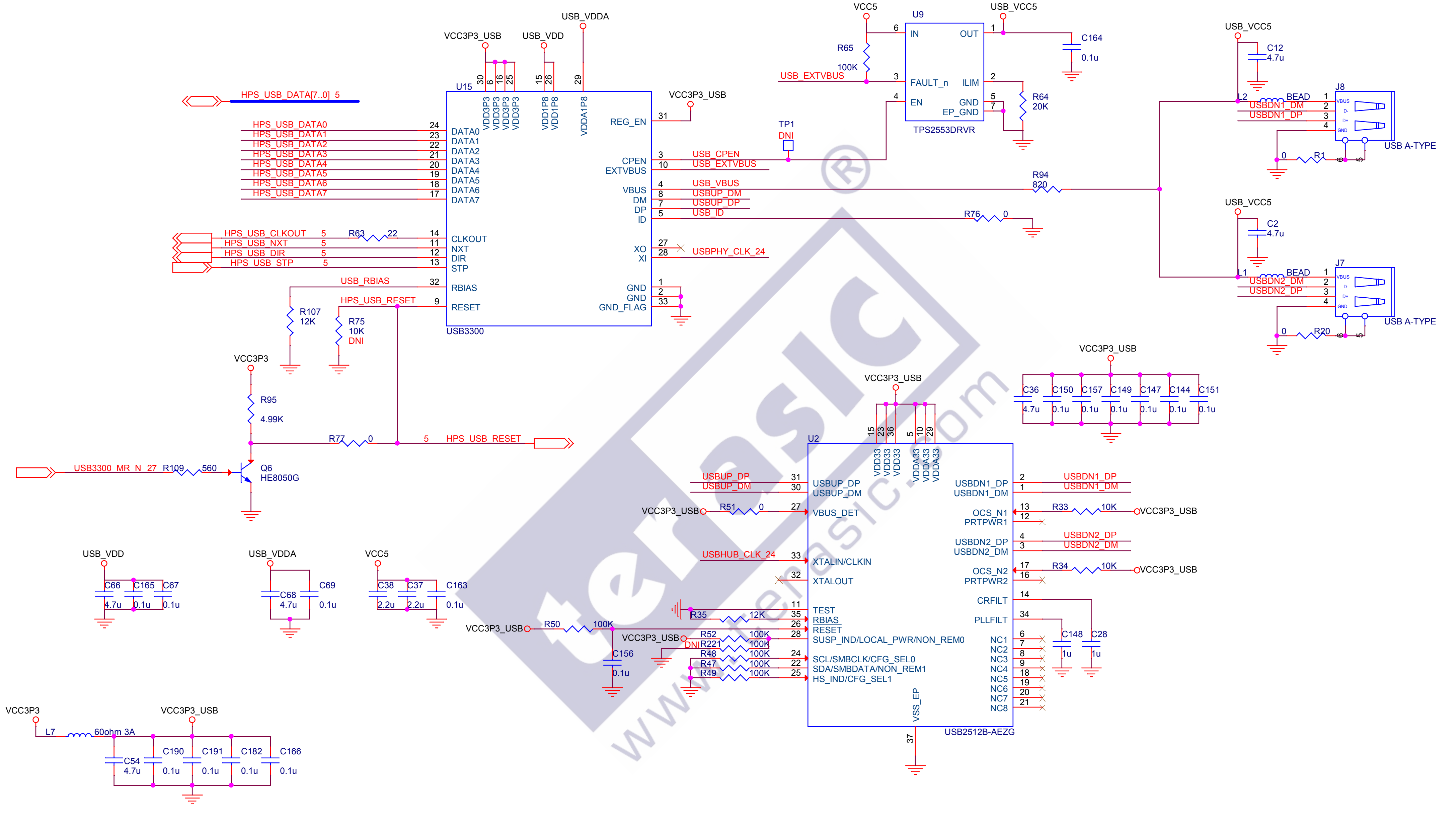
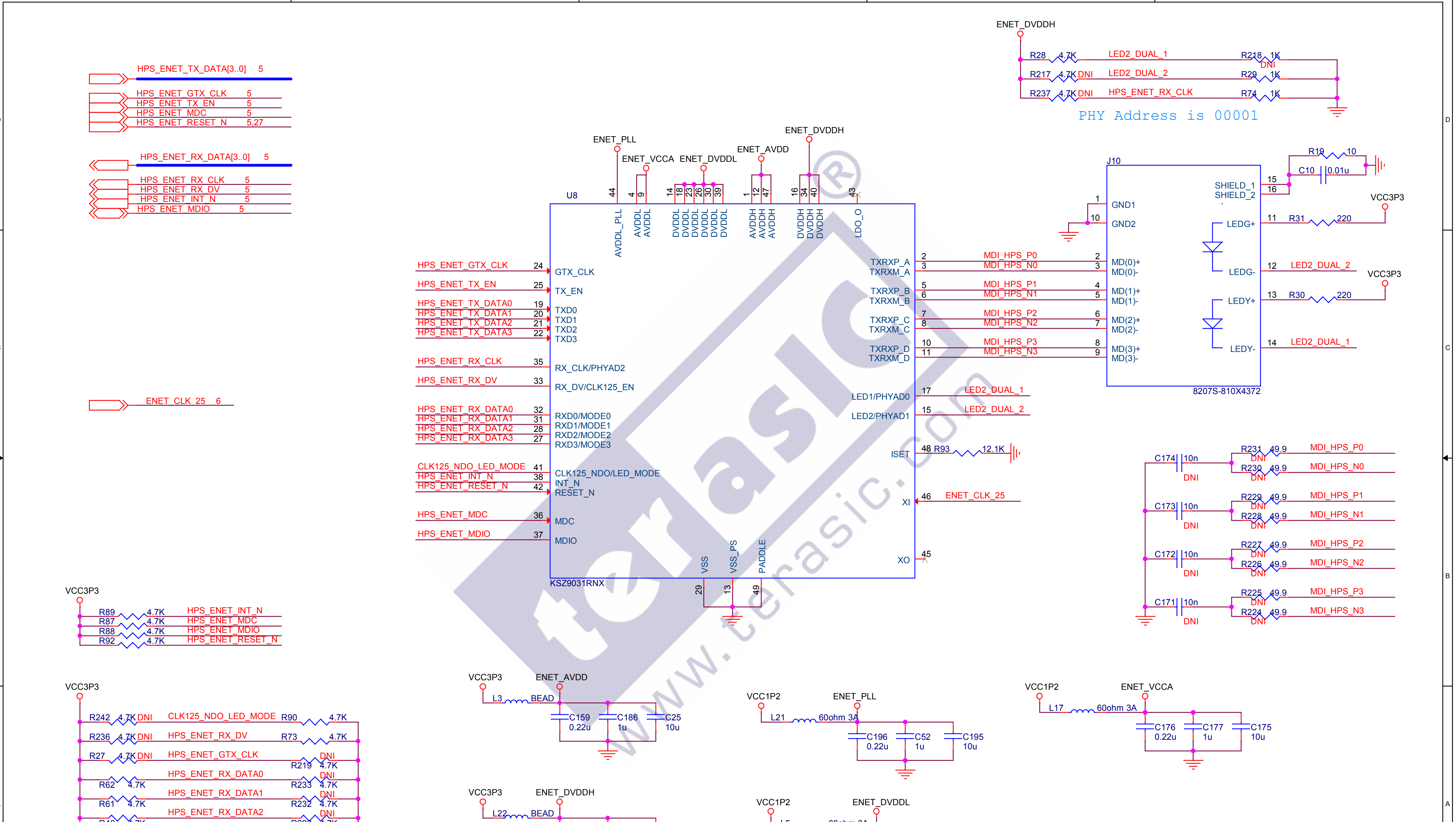
Title: **DE10-Standard Board**

Size	Document Number	Rev
B	ADC, PS2, IR Tx, IR Rx	D

Date: Thursday, November 03, 2022 — Sheet 22 of 33



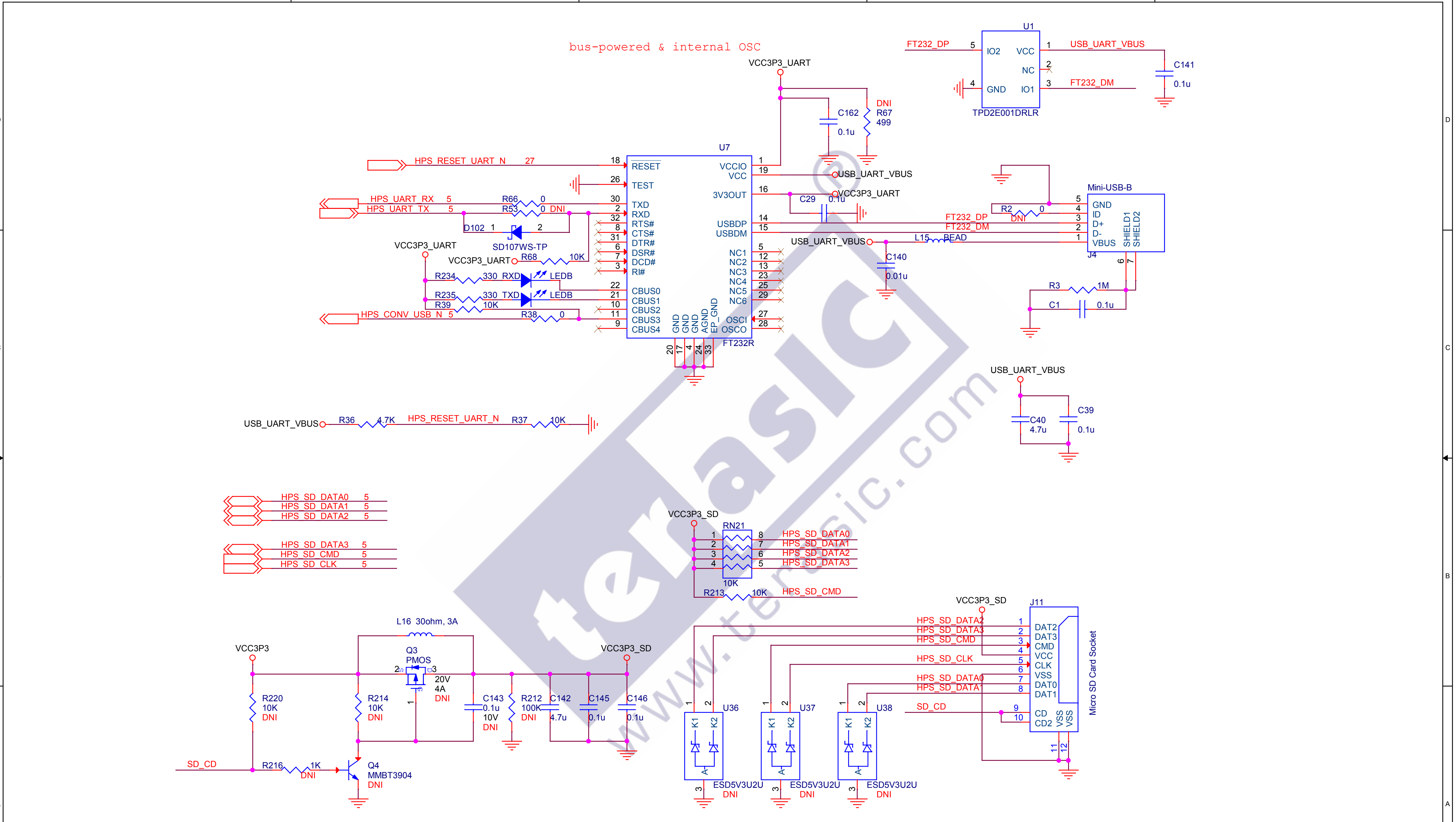
Title		
DE10-Standard Board		
Size: B	Document Number: 2-port USB Host	Rev: D
Date: Thursday, November 03, 2022	Sheet 23 of 33	



Title: **DE10-Standard Board**

Size	Document Number	Rev
B	1 Gigabit Ethernet	D

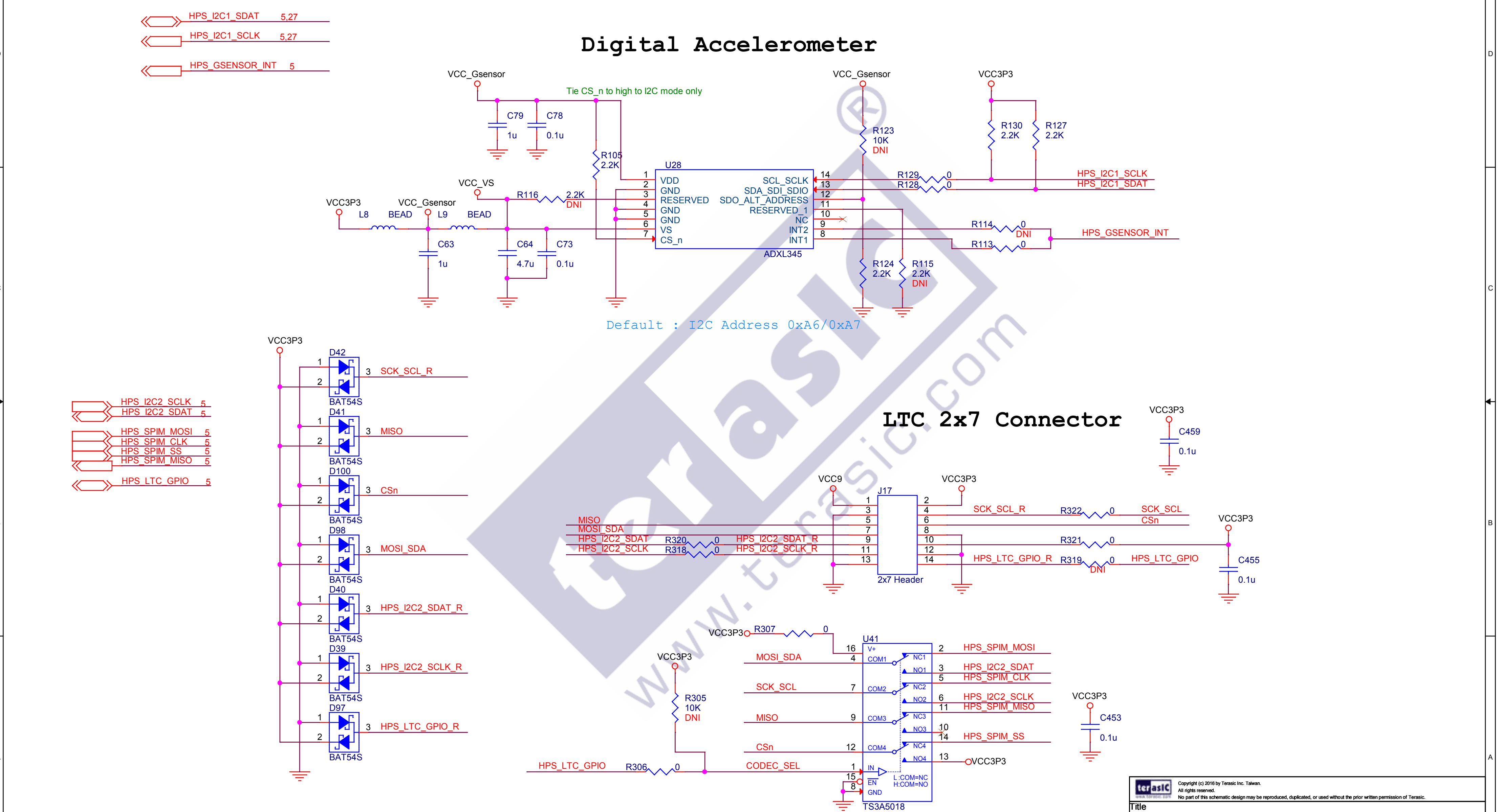
Date: Thursday, November 03, 2022 Sheet 24 of 33



Title: **DE10-Standard Board**

Size	Document Number	Rev
B	UART to USB, SD CARD	D

Date: Thursday, November 03, 2022



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Title	DE10-Standard Board	
Size: B	Document Number: Accelerometer, LTC Connector	Rev: D
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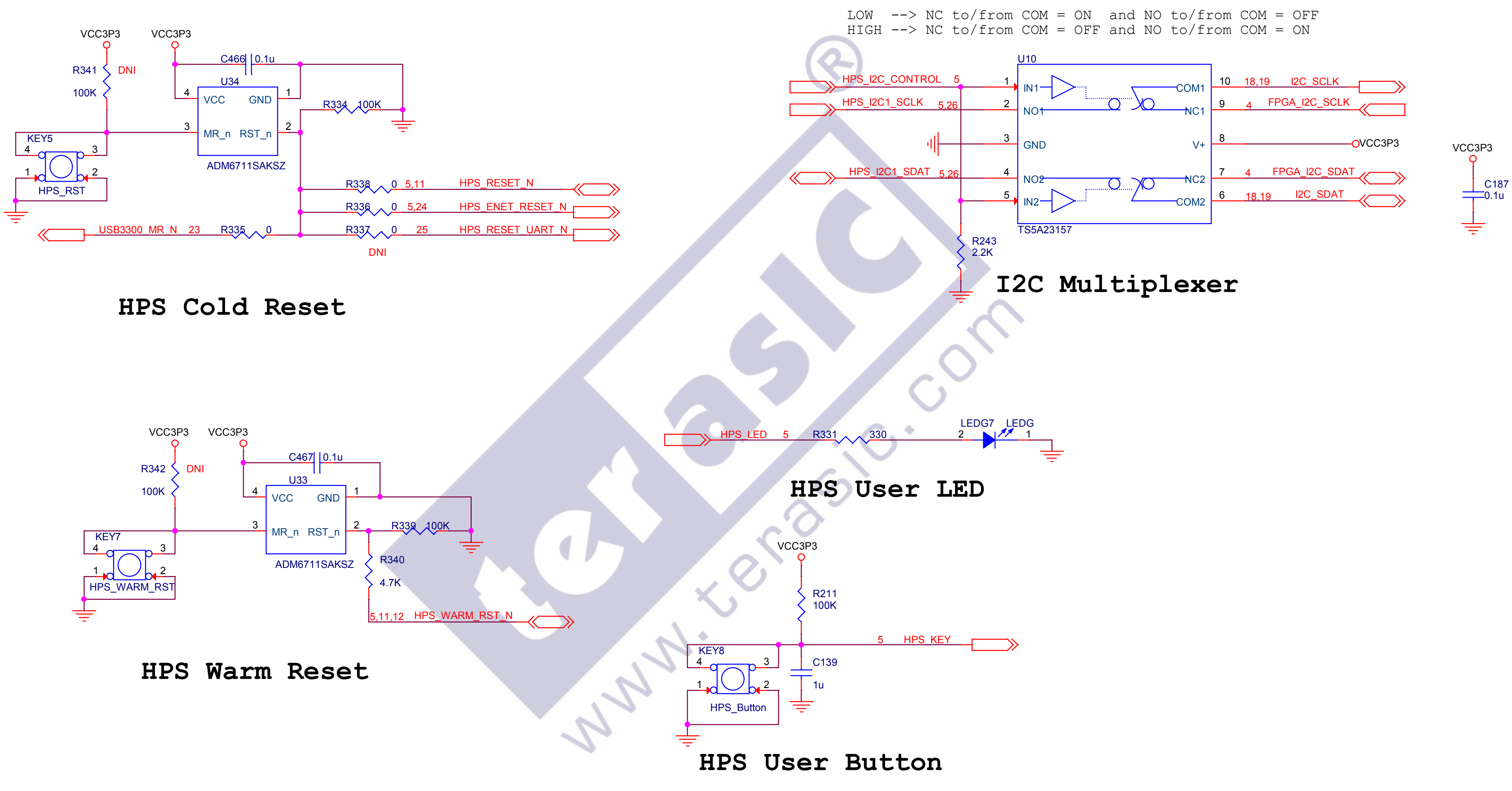
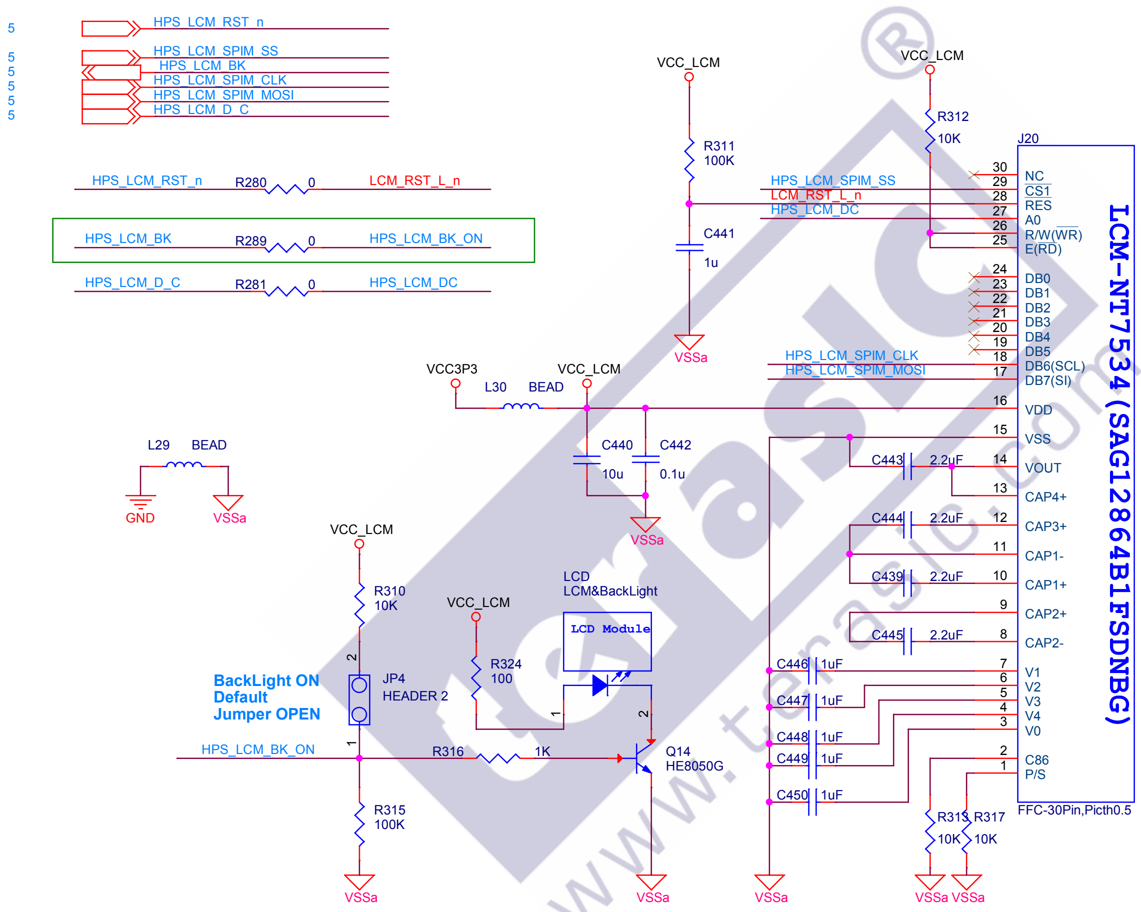
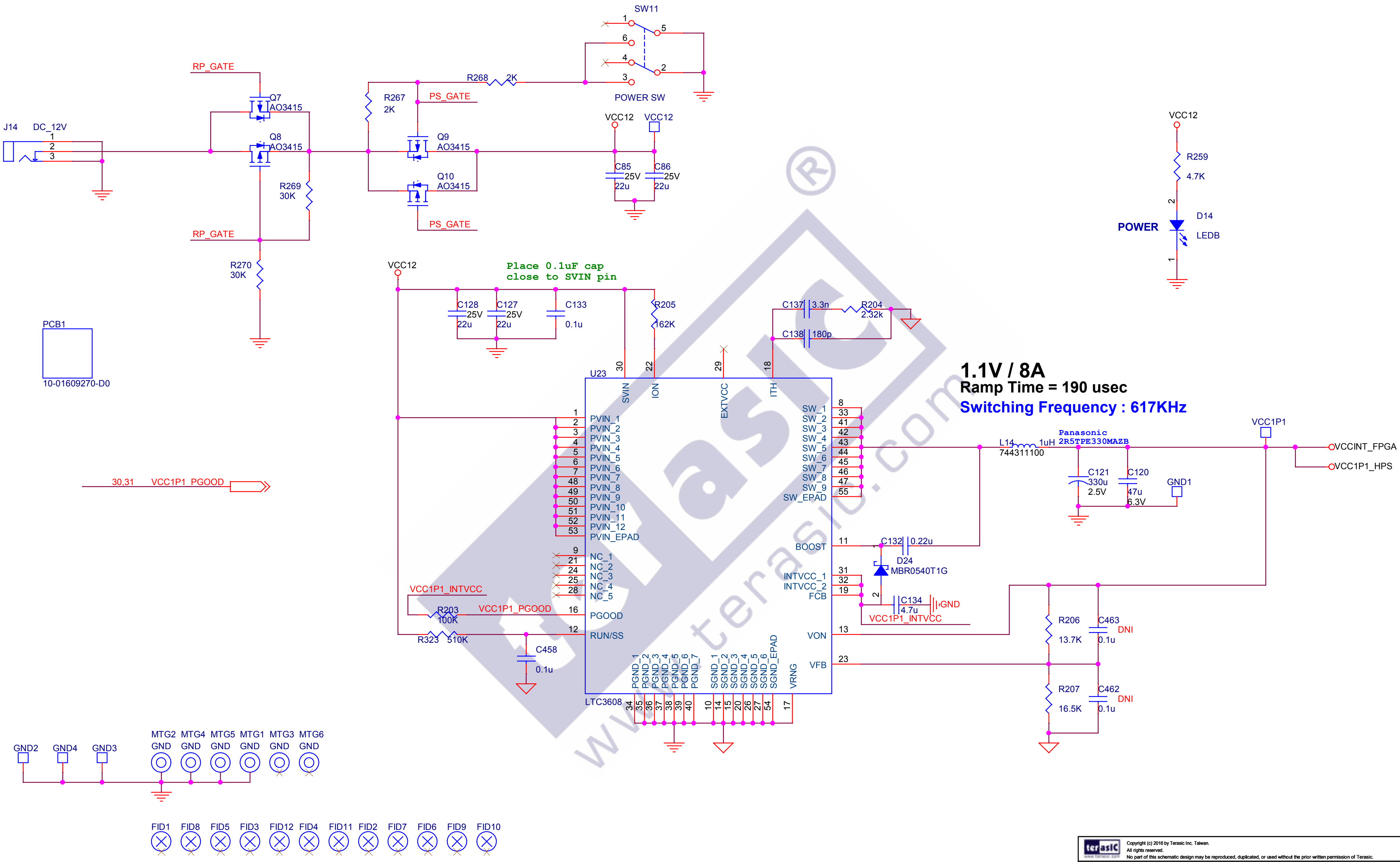


Title	DE10-Standard Board	
Size B	Document Number I2C Multiplexer, HPS BUTTON, HPS LED	Rev D
Date: Thursday, November 03, 2022	Sheet 27 of 33	



Title	DE10-Standard Board	
Size B	Document Number LCD	Rev D
Date: Thursday, November 03, 2022	Sheet 28 of 33	



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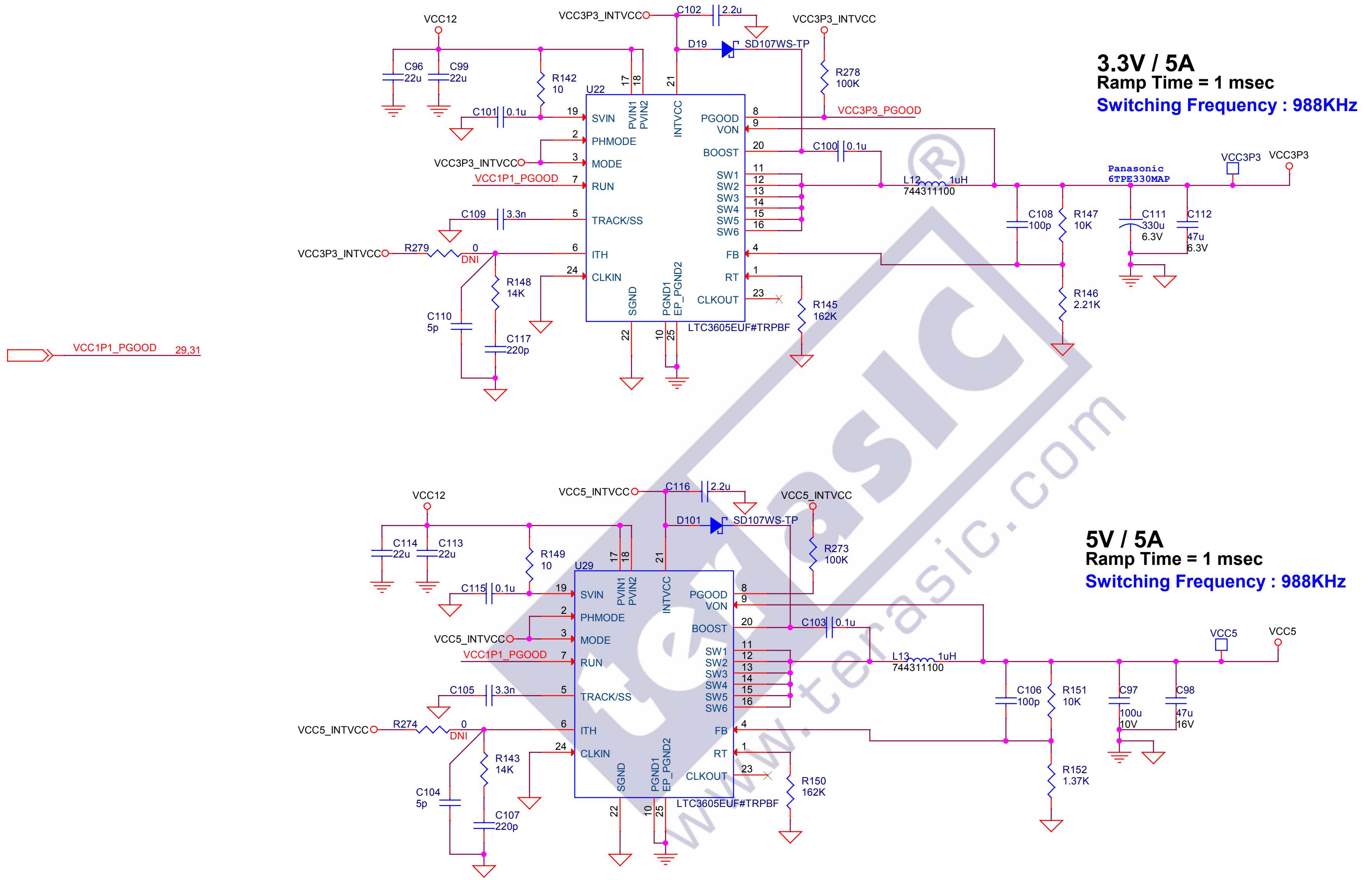
Title	DE10-Standard Board	
Size B	Document Number Power - 1.1V	Rev D
Date: Thursday, November 03, 2022	Sheet 29 of 33	



Title	DE10-Standard Board	
Size B	Document Number: Power - 5V, 3.3V	Rev D
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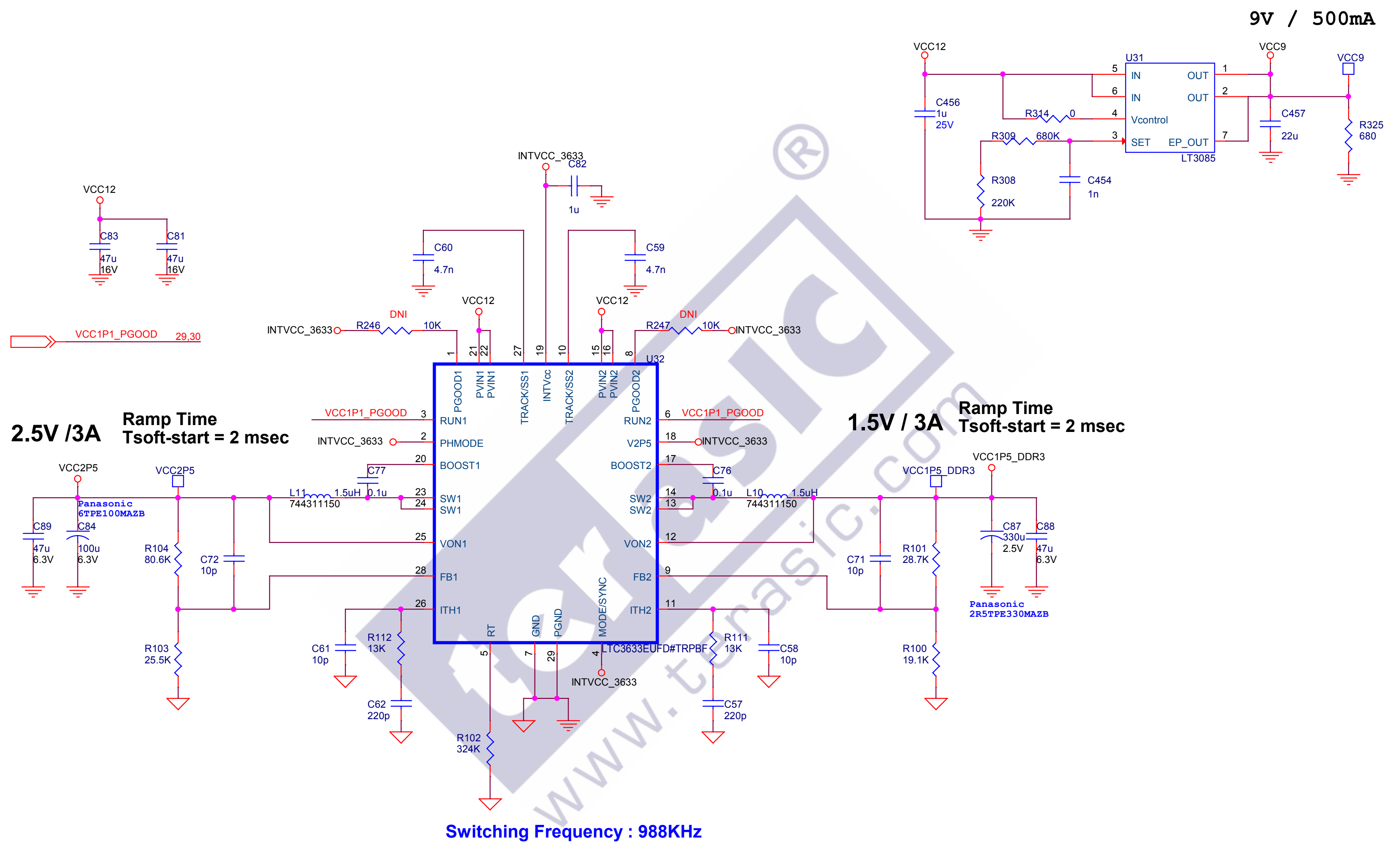
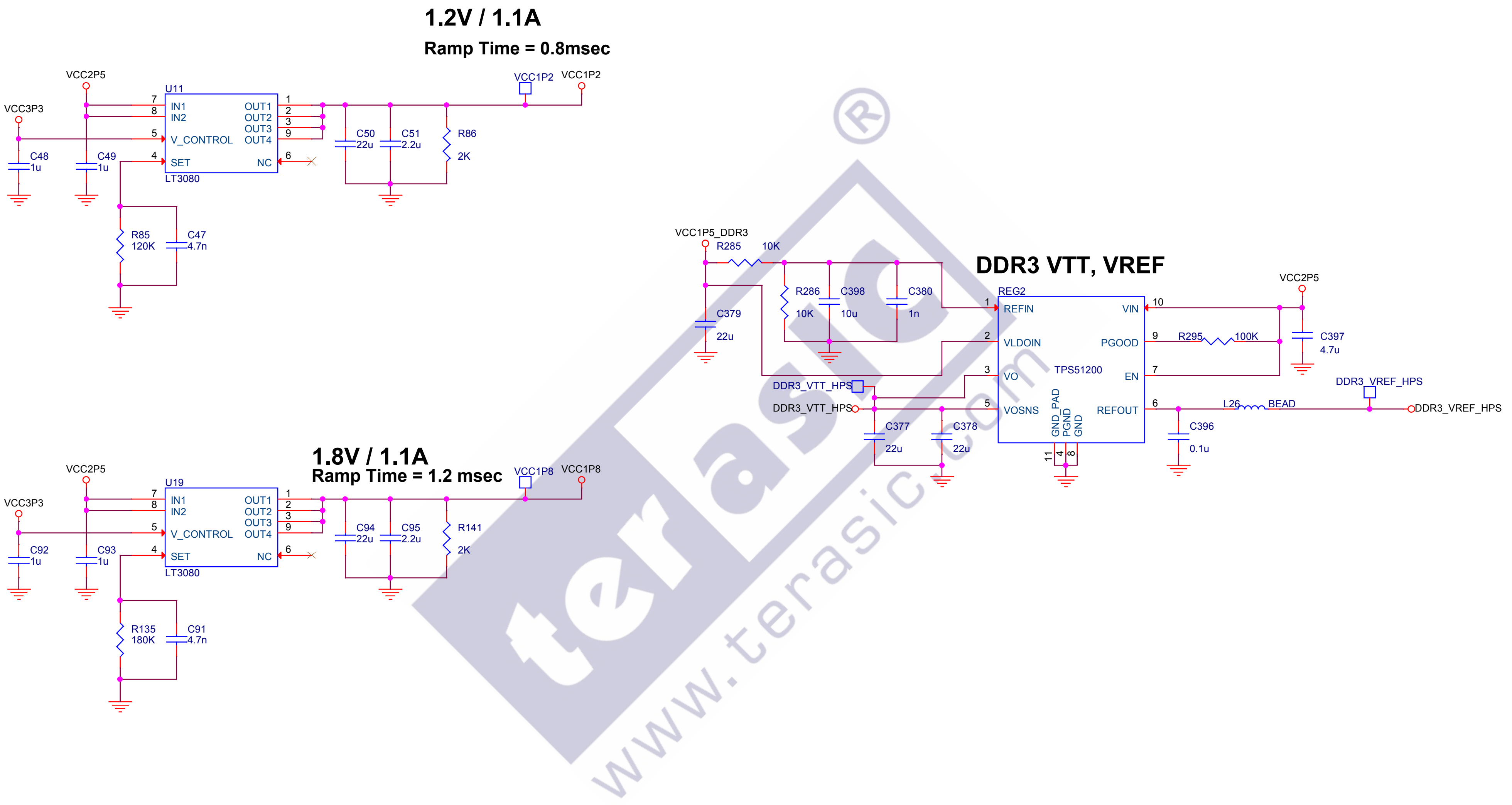


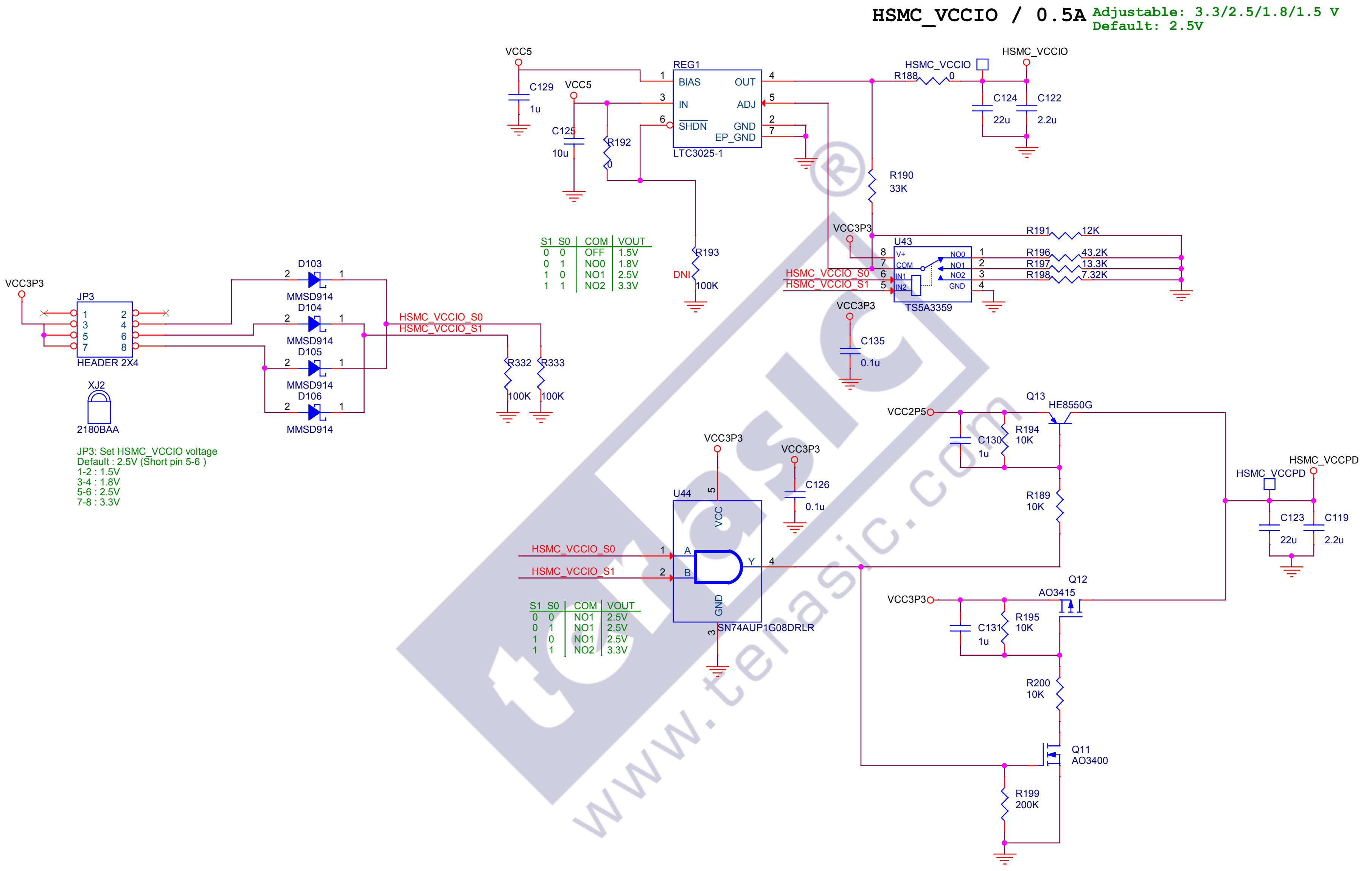
Title
DE10-Standard Board

Size	Document Number	Rev
B	Power - 9V, 2.5V, 1.5V	D

Date: Thursday, November 03, 2022 Sheet 31 of 33



Title		
DE10-Standard Board		
Size: B	Document Number: Power - 1.2V, 1.8V, DDR3 VREF, DDR3 VTT	Rev: D
Date: Thursday, November 03, 2022	Sheet 32 of 33	



S1	S0	COM	VOUT
0	0	OFF	1.5V
0	1	NO0	1.8V
1	0	NO1	2.5V
1	1	NO2	3.3V

S1	S0	COM	VOUT
0	0	NO1	2.5V
0	1	NO1	2.5V
1	0	NO1	2.5V
1	1	NO2	3.3V

Title
DE10-Standard Board

Size B | Document Number: Power - VCCIO_HSMC & HSMC_VCCPD | Rev D

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